

UNIT-I: FINITE AUTOMATA

1. Analytically, a finite automaton can be represented by a _____ tuples:
 - a. 6
 - b. 4
 - c. 5
 - d. 7
2. Final state(s) may not be a part of the given states of a finite automaton.
 - a. True
 - b. False
 - c. Both
 - d. None of These
3. Transition Systems is the other name given to:
 - a. Transition table
 - b. Transition diagram
 - c. States of automata
 - d. Automata as a whole
4. Implicitly, $\delta(q, \wedge) = \underline{\hspace{2cm}}$
 - a. q
 - b. $\wedge$
 - c. Initial state
 - d. Final state
5. For any transition function δ and for any two input strings x and y, $\delta(q, xy) = \underline{\hspace{2cm}}$
 - a. $\delta(q, x).\delta(q, y)$
 - b. $\delta(q, yx)$
 - c. $\delta(q, x) \times \delta(q, y)$
 - d. $\delta(\delta(q, x), y)$
6. DFA stands for:
 - a. Deterministic Final Automata
 - b. Deterministic in-Finite Automata
 - c. Determined Finite Automata
 - d. Deterministic Finite Automata
7. The function $Z(t) = \lambda(q(t), x(t))$ represents:
 - a. Moore Machine
 - b. Deterministic Finite Automata
 - c. Mealy Machine
 - d. Non-Deterministic Finite Automata
8. The function $Z(t) = \lambda(q(t))$ represents:
 - a. Moore Machine
 - b. Deterministic Finite Automata
 - c. Mealy Machine
 - d. Non-Deterministic Finite Automata

9. In a transition table, initial state is depicted by which of the following symbol?
- Arrow pointing to a state($\rightarrow$)
 - Circle over a state
 - Concentric circles over a state
 - Underlined state
10. NDFA stands for:
- Non-Deterministic Final Automata
 - Non-Finite Deterministic Automata
 - Non-Deterministic Finite Automata
 - Non-Determined Finite Automata
11. Final state in a Transition Diagram is depicted by:
- Arrow pointing to a state($\rightarrow$)
 - Circle over a state
 - Concentric circles over a state
 - Underlined state
12. Minimization of Finite automata means:
- Converting an NDFA into a DFA
 - Reducing number of states in a finite automata
 - Creating finite automata with minimum number of final states
 - Reducing number of transitions in a finite automata

Answer keys: Unit-I

- | | | | |
|------|------|------|-------|
| 1. c | 4. a | 7. c | 10. c |
| 2. b | 5. d | 8. a | 11. c |
| 3. b | 6. d | 9. a | 12. b |

UNIT-II: REGULAR EXPRESSIONS AND REGULAR SETS

1. Machine format of Regular Expression is
 - a. Finite Automata
 - b. Push Down Automata
 - c. Turing Machine
 - d. All of the above

2. Regular Expression is accepted by
 - a. Finite Automata
 - b. Push Down Automata
 - c. Turing Machine
 - d. All of the above

3. The language of all words with atleast 2 a's can be described by the Regular Expression
 - a. $a(ab)^*a$
 - b. $(a + b)^*ab^*a(a + b)^*$
 - c. $b^*ab^*a(a + b)^*$
 - d. All of the above

4. Which type of language is Regular Expression?
 - a. Type 0
 - b. Type 1
 - c. Type 2
 - d. Type 3.

5. Which of the following Regular Expression over $\{0,1\}$ denotes set of all strings not containing 100 as substring?
 - a. $(1 + 0)^*0^*$
 - b. 0^*1010^*
 - c. 0^*1^*01
 - d. All of the above.

6. $R^+ = ?$
 - a. R
 - b. R^*
 - c. $R +$
 - d. R^+

7. If L = language of words containing even number of a's. The corresponding Regular Expression is:
 - a. $(a+b)^*aa(a+b)^*$

- b. $(b+ab^*a)^*$
- c. $a+bb^*aab^*a$
- d. $(a+b)^*ab(a+b)^*$

8. The language that can be expressed by any regular expression is called a Non-regular language.

- a. True
- b. False

9. Arden's theorem states that if $R = Q + RP$, then _____ is its unique solution.

- a. $R = QP^*$
- b. $R = PQ^*$
- c. $R = P^*Q^*$
- d. $R = Q^*P^*$

10. The regular expression described by following automata is,



- a. 01^*01^*
- b. $(01)^*$
- c. 0^*1^*
- d. 1^*0^*

11. Which of the following are the examples of non-regular languages.

- a. PALINDROME and PRIME
- b. PALINDROME and EVEN-EVEN
- c. EVEN-EVEN and PRIME
- d. FACTORIAL and SQUIRE

12. Languages are proved to be regular or non-regular using pumping lemma.

- a. True
- b. False

Answer keys: Unit-II

1. a

2. a

3. d

4. d

5. c
6. b

7. b
8. b

9. a
10. c

11. a
12. a

UNIT-III: FORMAL LANGUAGES AND REGULAR GRAMMARS

1. Noam Chomsky gave a mathematical model of a grammar in:
 - a. 1966
 - b. 1956
 - c. 1946
 - d. 1976
2. A grammar is also called a _____structured grammar.
 - a. Phrase
 - b. Pre
 - c. Parse
 - d. Class
3. The elements of V_N in a grammar are called:
 - a. Terminals
 - b. Starting Symbols
 - c. Variables
 - d. Production rules
4. For any grammar $G = (V_N, \Sigma, P, S)$, $V_N \cap \Sigma =$ _____.
 - a. Σ
 - b. $V_N \cup \Sigma$
 - c. $\emptyset$
 - d. V_N
5. S in a grammar $G = (V_N, \Sigma, P, S)$ is called the:
 - a. Stop symbol
 - b. Special symbol
 - c. Syntax symbol
 - d. Start symbol
6. Grammatical rules which involve the meaning of words are called:
 - a. Semantics
 - b. Syntactic
 - c. Both a and b
 - d. None of given
7. Grammatical rules which do not involve the meaning of words are called:
 - a. Semantics
 - b. Syntactic
 - c. Both a and b
 - d. None of given
8. The symbols in a grammar that can't be replaced by anything are called:
 - a. Productions
 - b. Terminals

- c. Non-terminals
 - d. All of above
9. The symbols in a grammar that must be replaced by other things are called:
- a. Productions
 - b. Terminals
 - c. Non-terminals
 - d. None of given
10. The grammatical rules are often called:
- a. Productions
 - b. Terminals
 - c. Non-terminals
 - d. None of given
11. The terminals are designated by _____ letters, while the non-terminals are designated by _____ letters.
- a. Capital, bold
 - b. Small, capital
 - c. Capital, small
 - d. Small, bold
12. L(G) is an acronym used for:
- a. Leaves of derivation tree for a grammar G
 - b. Length of a grammar G
 - c. Left hand symbols of all productions in G
 - d. Language generated by a grammar G

Answer keys: Unit-III

- 1. b
- 2. a
- 3. c
- 4. c
- 5. d
- 6. a
- 7. b
- 8. b
- 9. c
- 10. a
- 11. b
- 12. d

UNIT-I: FINITE AUTOMATA

13. While minimizing a finite automata, we generally create the:
 - a. Equivalence states
 - b. Equivalence transitions
 - c. Equivalence classes
 - d. Equivalence groups
14. A Moore machine is a six-tuple $(Q, \Sigma, \Delta, \delta, \lambda, q_0)$ where Δ is the:
 - a. Output alphabet
 - b. Input alphabet
 - c. Transition function
 - d. Output function
15. A finite automata may contain more than one final states:
 - a. True
 - b. False
16. The number of columns in a transition table for a finite automata are:
 - a. Three
 - b. One less than the number of input symbols
 - c. Same as the number of input symbols
 - d. One more than the number of input symbols
17. The transition function that maps $Q \times \Sigma^*$ into Q is called:
 - a. Direct transition function
 - b. Indirect transition function
 - c. Simple transition function
 - d. Directed transition function
18. When next state of an automata at any instant of time is determined by the present state and the present input, then it is called a:
 - a. State relation
 - b. Output relation
 - c. Set of states
 - d. Set of input symbols
19. A string x is accepted by a finite automata if $\delta(q_0, x) = q$; where q is:
 - a. Initial state
 - b. Final state
 - c. Any state
 - d. None of the above
20. Q is a set of states in finite automata. What holds true for Q :
 - a. Q may contain any number of states
 - b. Q contains finite number of states
 - c. Q does not contain any final state

- d. Q is not a set of states.
- 21. Determine the Equivalent Machine that takes present state and output
 - a. Moore
 - b. Mealy
 - c. Both
 - d. None
- 22. How many tuples are there in Finite Automata
 - a. 4
 - b. 5
 - c. 6
 - d. 7
- 23. Minimized automata should be
 - a. DFA
 - b. NFA
 - c. Both
 - d. None
- 24. Finite Automata can deal with which language
 - a. Regular Languages
 - b. Context Free
 - c. Context Sensitive
 - d. Recursive Enumerable

Answer keys: Unit-I

- 1. C
- 2. A
- 3. A
- 4. D
- 5. B
- 6. A
- 7. B
- 8. B
- 9. B
- 10. B
- 11. A
- 12. A

UNIT-II: REGULAR EXPRESSIONS AND REGULAR SETS

1. In pumping lemma theorem (xy^iz) the range of i is

- a. $i = \dots -3, -2, -1, 1, 2, 3, 4, \dots$
- b. $i = 0, 1, 2, 3, 4, \dots$
- c. $i = \dots -3, -2, -1, 0, 1, 2, 3, 4, \dots$
- d. $i = 1, 2, 3, 4, \dots$

2. To examine whether a certain FA accepts any words, it is required to seek the paths from _____ state.

- a. Final to initial

- b. Final to final
- c. Initial to final
- d. Initial to initial

3. If $\Sigma = \{a,b\}$ and given productions are

$$S \rightarrow XaaX$$

$$X \rightarrow aX \mid bX \mid \Lambda$$

Then the above grammar defines the language expressed by _____ regular expression

- a. $(a+b)^*aa(a+b)^*$
- b. $(a+b)^*a(a+b)^*a$
- c. $(a+b)^*aa(a+b)^*aa$
- d. $(a+b)^*aba+b)^*$

4. A regular expression can be defined as a language or string accepted by a finite automata.

- a. True
- b. False

5. If w BELONGS TO $(a, b)^*$ satisfy $abw = wab$, then (w) is

- a. Null
- b. Odd
- c. Even
- d. None of these

6. Identity: $\Lambda + R = \underline{\hspace{2cm}}$

- a. Λ
- b. RR
- c. R^*
- d. R

7. Identity: $R + R = \underline{\hspace{2cm}}$

- a. $2R$
- b. RR
- c. R
- d. R^*

8. Identity: $(R^*)^* = \underline{\hspace{2cm}}$

- a. R^{**}
- b. R^*R^*
- c. R^*
- d. R

9. The languages which can not accept PDA_____

- A. REGULAR LANGUAGES
- B. CONTEXT FREE
- C. CONTEXT SENSITIVE
- D. NONE.

10. CONVERSION OF NFA TO DFA WILL LEAD TO HOW MANY INITIAL STATES

- A. 2
- B. 1
- C. NONE
- D. BOTH

11. How many strings of length less than 4 contains the language described by the regular expression $(x+y)^*y(a+ab)^*$?

- a) 7
- b) 10
- c) 12
- d) 11

12 . A language is regular if and only if

- a) accepted by DFA
- b) accepted by PDA
- c) accepted by LBA
- d) accepted by Turing machine

Answer keys: Unit-II

- 1. D
- 2. C
- 3. A
- 4. A
- 5. C
- 6. D
- 7. C

- 8. C
- 9. A
- 10. B
- 11. D
- 12. A

UNIT-III: FORMAL LANGUAGES AND REGULAR GRAMMARS

13. The highest type of grammar under Chomsky classification is:
- Type-3
 - Type-4
 - Type-2
 - Type-0
14. Type-1 grammars are also called:
- Phrase structure grammars
 - Context free grammars
 - Context sensitive grammars
 - Regular grammars
15. A production of the form $A \rightarrow a$ and $A \rightarrow aB$ is called a _____ production.
- Type-0
 - Type-3
 - Type-1
 - Type-2
16. The machine counterpart of Context Free Grammars is:
- Finite automata
 - Pushdown automata
 - Turing machine
 - Linear bounded automata
17. The machine counterpart of Regular Grammars is:
- Finite automata
 - Pushdown automata
 - Turing machine
 - Linear bounded automata
18. _____ is the corresponding automata for Context Sensitive languages:
- Linear bounded automata
 - Turing machine
 - Finite automata
 - Pushdown automata
19. The symbol used for step by step derivation of a grammar is:
- $\rightarrow$
 - $\Rightarrow$
 - $\leftrightarrow$
 - $\rightleftarrows$
 -
20. Language generated by grammar $G = (\{S, C\}, \{a, b\}, P, S)$ where P consists of $S \rightarrow aCa, C \rightarrow aCa \mid b$ is:
- ab^na

- b. $a^n b^n a^n$
 - c. $ab^n a^n$
 - d. $a^n b a^n$
21. Regular grammar is
- a) Context free grammar
 - b) Non context free grammar
 - c) English grammar
 - d) None of the mentioned
22. Regular expression are
- a) Type 0 language
 - b) Type 1 language
 - c) Type 2 language
 - d) Type 3 language
23. L and $\sim L$ are recursive enumerable then L is
- a) Regular
 - b) Context free
 - c) Context sensitive
 - d) Recursive
24. . Regular expressions are closed under
- a) Union
 - b) Intersection
 - c) Kleene star
 - d) All of the mentioned

Answer keys: Unit-III

13. a
14. c
15. b
16. b
17. a
18. a

19. b
20. d

21. a
22. a

23. d
24. d

Unit 4 :CONTEXT FREE GRAMMARS + SIMPLIFICATION OF CONTEXT- FREE GRAMMARS

1. The production of the form $\text{no terminal} \rightarrow \Lambda$ is said to be null production.
 - a. TRUE
 - b. FALSE
2. A production is called null able production if it is of the form $N \rightarrow \Lambda$
 - a. TRUE
 - b. FALSE
3. The productions of the form $\text{nonterminal} \rightarrow \text{one nonterminal}$, is called _____
 - a. Null production
 - b. Unit production
 - c. Null able production
 - d. None of given
4. CNF is stands for
 - a. Context Normal Form
 - b. Complete Normal Form
 - c. Chomsky Normal Form
 - d. Compared Null Form
5. GNF stands for
 - a. Gray Normal Form
 - b. Greibach Normal Form
 - c. Graphite Normal Form
 - d. Greatest Normal Form
6. The derivation of the word W generated by a CFG, such that at each step,a production is applied to the left most nonterminal in the working string is said to be
 - a. Left most terminal
 - b. Left most deviation
 - c. None of these
 - d. Both a and b
7. The process of finding the derivation of word generated by particular grammar is called :
 - a. PLUS TIMING
 - b. Parsing
 - c. HALT
 - d. All of above
8. A production in CFG consists of:
 - a. One terminal
 - b. More than one terminal
 - c. One non-terminal
 - d. Terminals and non-terminals
9. The entity which generate Language is termed as:
 - a) Automata
 - b) Tokens

- c) Grammar
d) Data
10. The Grammar can be defined as: $G=(V, \Sigma, p, S)$
In the given definition, what does S represents?
a) Accepting State
b) Starting Variable
c) Sensitive Grammar
d) None of these
11. Which of the expression is appropriate?
For production p: $a \rightarrow b$ where $a \in V$ and $b \in \underline{\hspace{2cm}}$
a) V
b) S
c) $(V+\Sigma)^*$
d) $V+\Sigma$
12. For $S \rightarrow 0S1 \mid \epsilon$ for $\Sigma=\{0,1\}^*$, which of the following is wrong for the language produced?
a) Non regular language
b) $0^n 1^n \mid n \geq 0$
c) $0^n 1^n \mid n \geq 1$
d) None of the mentioned

Answer Keys: Unit 4

1. a
2. a

3. b
4. c
5. b
6. b
7. b

8. d
9. C
10. B
11. C
12. d

Unit V: PUSHDOWN AUTOMATA AND PARSING

1. For a given Pushdown automata, we can construct an equivalent context free grammar:
 - a. True
 - b. False

2. In _____ parsing, we attempt to construct a derivation of the input string, starting from the root and ending in the given input string.
 - a. Top-down
 - b. Top-Up
 - c. Bottom-Up
 - d. Bottom-Down

3. In _____ parsing, we attempt to construct a derivation from the given input string to the top ie, the root with label S.
 - a. Top-down
 - b. Top-Up
 - c. Bottom-Up
 - d. Bottom-Down

4. In LL(k), the first L stands for _____:
 - a. Left
 - b. Look from right
 - c. Lost symbols
 - d. Look ahead

5. In LL(k), the second L stands for _____:
 - a. Look from right
 - b. Left
 - c. Lost symbols
 - d. Look ahead

6. In LL(k), k represents:
 - a. Number of symbols to look behind
 - b. Number of symbols to look ahead
 - c. Number of input symbols in the string
 - d. Number of symbols towards left of current symbol

7. In LL(k), the value of k can also be negative.
 - a. True
 - b. False

8. In order to do top-down parsing for a general string in L(G), we prepare a table called:
 - a. Transition table
 - b. Parse Table
 - c. Top-down Table
 - d. Production Table

9. The transition a Push down automaton makes is additionally dependent upon the:
 - a) stack
 - b) input tape
 - c) terminals
 - d) none of the mentioned

10. With reference of a DPDA, which among the following do we perform from the start state with an empty stack?
 - a) process the whole string

- b) end in final state
- c) end with an empty stack
- d) all of the mentioned

11. If the PDA does not stop on an accepting state and the stack is not empty, the string is:

- a) rejected
- b) goes into loop forever
- c) both (a) and (b)
- d) none of the mentioned

12. A language accepted by Deterministic Push down automata is closed under which of the following?

- a) Complement
- b) Union
- c) Both (a) and (b)
- d) None of the mentioned

Answer keys: Unit-V

- | | |
|------|-------|
| 1. a | 7. b |
| 2. a | 8. b |
| 3. c | 9. a |
| 4. d | 10. d |
| 5. b | 11. a |
| 6. b | 12. a |

Unit V1: TURING MACHINES AND COMPLEXITY

1. A Turing machine can be:

- a. Multi-Tape Turing Machine
- b. Multi-head Turing Machine
- c. Universal Turing machine
- d. All of the above

2. A Turing machine can only be deterministic.

- a. True
- b. False

3. A Turing machine can be a Multi-Track Turing machine.

- a. True
- b. False

4. Who is known as the father of Computer Science?

- a. S. C. Kleene
- b. Alanzo Church
- c. Charles Babbage
- d. Alan Turing

5. A _____ is a theoretical computing machine invented by Alan Turing (1937) to serve as an idealized model for mathematical calculation.

- a. Linear Bounded Automata
- b. Turing Machine
- c. Finite Automata
- d. Pushdown Automata

6. A Turing machine can run forever, enter a loop, or reach a particular state or set of conditions at which it is prescribed to halt.

- a. True
- b. False

7. In computer science, a _____ is a Turing machine that can simulate an arbitrary Turing machine on arbitrary input.

- a. Universal Turing machine
- b. Deterministic Turing machine

- c. Non- Deterministic Turing machine
- d. Multi-Tape Turing machine

8. A _____ is the simplest form of a computer.

- a. Finite Automata
- b. Linear Bounded Automata
- c. Turing Machine
- d. Pushdown Automata

9. The machine accept the string by entering into hA or it can:

- a) explicitly reject x by entering into hR
- b) enter into an infinite loop
- c) Both (a) and (b)
- d) None of the mentioned

10. Which of the following can accept even palindrome over $\{a,b\}$

- a) Push down Automata
- b) Turing machine
- c) NFA
- d) All of the mentioned

11. If T1 and T2 are two turing machines. The composite can be represented using the expression:

- a) $T_1 T_2$
- b) $T_1 \cup T_2$
- c) $T_1 \times T_2$
- d) None of the mentioned

12. The number of states required to automate the last question i.e. $\{a,b\}^* \{aba\} \{a,b\}^*$ using finite automata:

- a) 4
- b) 3
- c) 5
- d) 6

Answer Keys: Unit V1:

- 1. d
- 2. b
- 3. a
- 4. d
- 5. b
- 6. a
- 7. a
- 8. c
- 9. c
- 10. c
- 11. a
- 12. a

Q1 Which of the following pairs of regular expressions are equivalent?

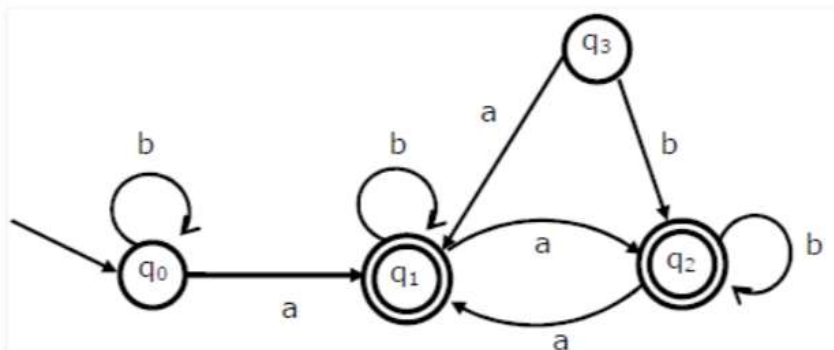
- a) $1(01)^*$ and $(10)^*1$
- b) $x(xx)^*$ and $(xx)^*x$
- c) $x^+ +$ and $x^+ x^+ (*+)$
- d) **All of the mentioned**

Q2 What is the Regular Expression Matching Zero or More Specific Characters

- a) x
- b) $\#$
- c) $*$
- d) $\&$

Q3

Consider the following Finite State Automaton. The language accepted by this automaton is given by the regular expression

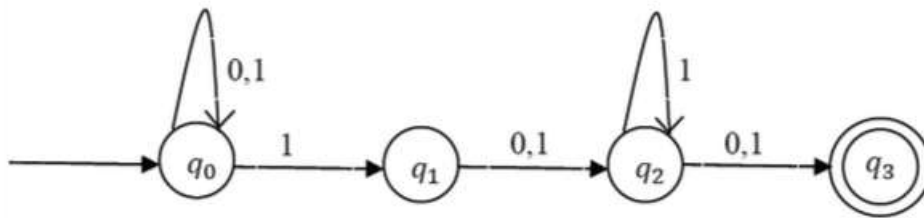


- (A) $b^*ab^*ab^*ab^*$ (B) $(a+b)^*$ (C) $b^*a(a+b)^*$ (D) $b^*ab^*ab^*$

Ans. C

Q4

Consider the finite automaton in the following figure.



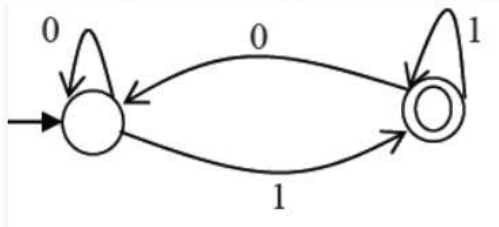
What is the set of reachable states for the input string 0011?

- (A) {q0, q1, q2}
- (B) {q0, q1}
- (C) {q0, q1, q2, q3}
- (D) {q3}

Ans. A

Q5

Which of the regular expressions given below represent the following DFA?



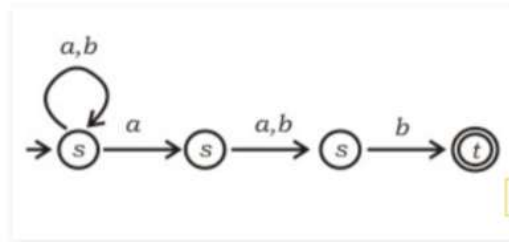
- I) $0^*1(1+00^*1)^*$
- II) $0^*1^*1+11^*0^*1$
- III) $(0+1)^*1$

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II, and III

Ans. B

Q6

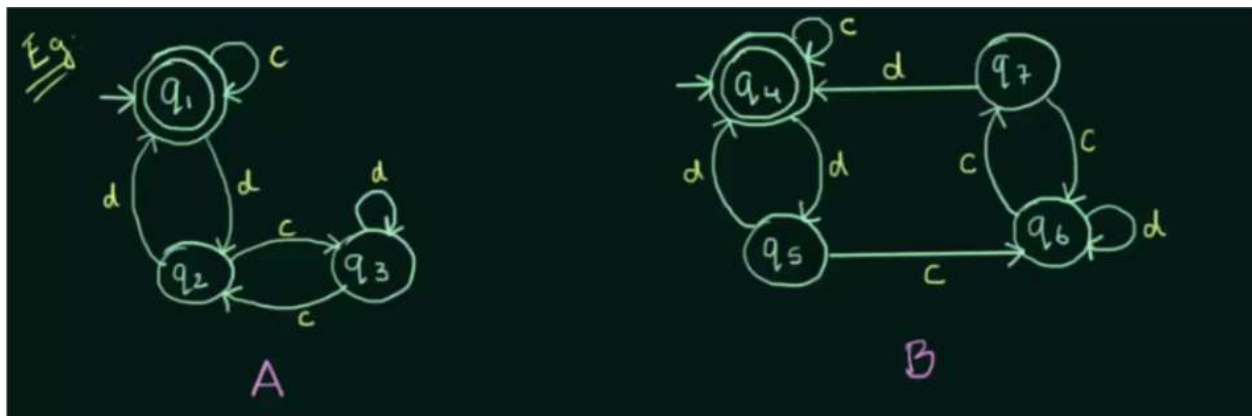
Which regular expression best describes the language accepted by the non-deterministic automaton below?



- (A) $(a + b)^* a(a + b)b$
- (B) $(abb)^*$
- (C) $(a + b)^* a(a + b)^* b(a + b)^*$
- (D) $(a + b)^*$

Ans. A

Q7 Consider the two automata



- A. A and B are Equivalent
- B. A and B are not Equivalent
- C. Cannot be defined
- D. None of these

Ans.A

Q8 Regular expression $(x/y)(x/y)$ denotes the set

- a. $\{xy, xy\}$
- b. $\{xx, xy, yx, yy\}$
- c. $\{x, y\}$
- d. $\{x, y, xy\}$

Ans.b

Q9 The regular expression with all strings of 0's and 1's with at least two consecutive 0's is:

- a. $1 + (10)^*$
- b. $(0+1)^*00(0+1)^*$
- c. $(0+1)^*011$
- d. $0^*1^*2^*$

Ans. B

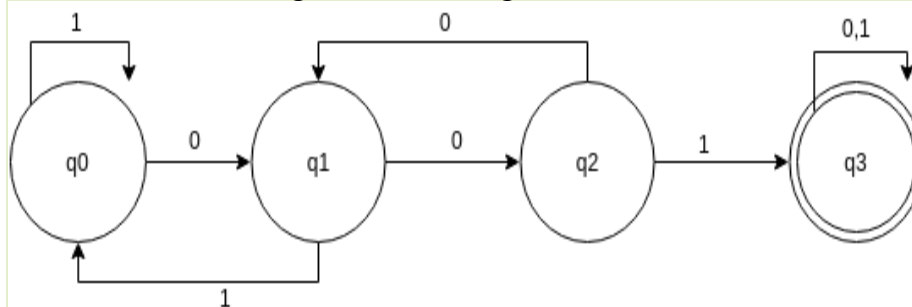
Q10 Find the pair of regular expressions that are equivalent

- a. $(0+1)^*$ and $(0^*+1^*)^*$
- b. $(0+1)^*$ and $(0+1^*)^*$
- c. $(0+10)^*$ and $(0^*+10)^*$
- d. All of the mentioned

Ans. D

Q11

Which of the following is same as the given DFA?

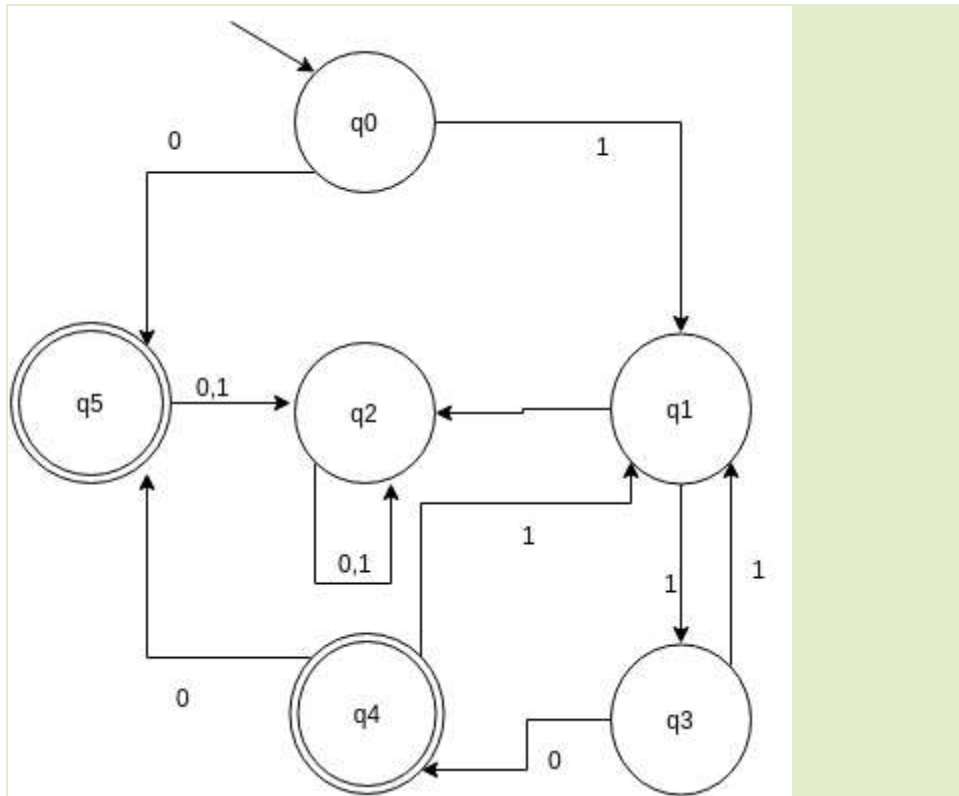


- a. $(0+1)^*001(0+1)^*$
- b. $1^*001(0+1)^*$
- c. $(01)^*(0+0+1)(01)^*$
- d. None of the mentioned

Ans. A

Q12

Which of the given regular expressions correspond to the automata shown?



- a. $(110+1)^*0$
- b. $(11+110)^*1$
- c. $(110+11)^*0$
- d. $(1+110)^*1$

Ans. C

Q13 Which among the following looks similar to the given expression?

$$((0+1). (0+1))^*$$

- a. $\{x \in \{0,1\}^* \mid x \text{ is all binary number with even length}\}$
- b. $\{x \in \{0,1\} \mid x \text{ is all binary number with even length}\}$
- c. $\{x \in \{0,1\}^* \mid x \text{ is all binary number with odd length}\}$
- d. $\{x \in \{0,1\} \mid x \text{ is all binary number with odd length}\}$

Ans. A

Q14 $(a + b^*c)$ most correctly represents:

- a. $(a + b)^*c$
- b. $(a) + ((b)^*.c)$
- c. $(a + (b^*)).c$
- d. $a + ((b^*).c)$

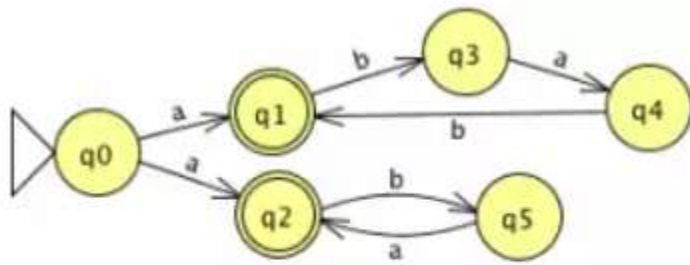
Ans. D

Q15

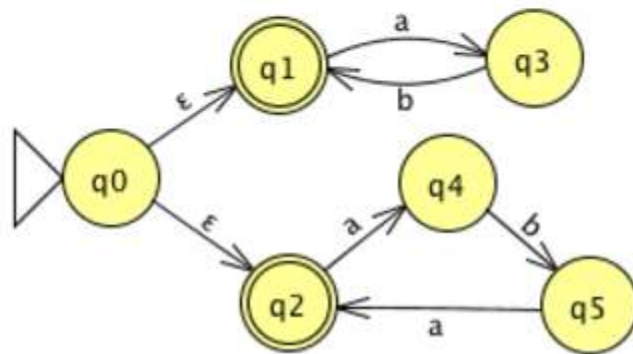
Which NDFA correctly represents the following RE

$a(bab)^* \cup a(ba)^*$

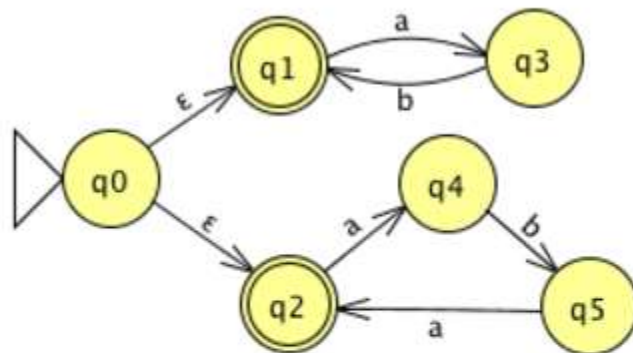
a)



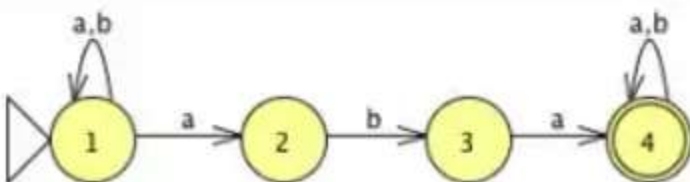
b)



c)

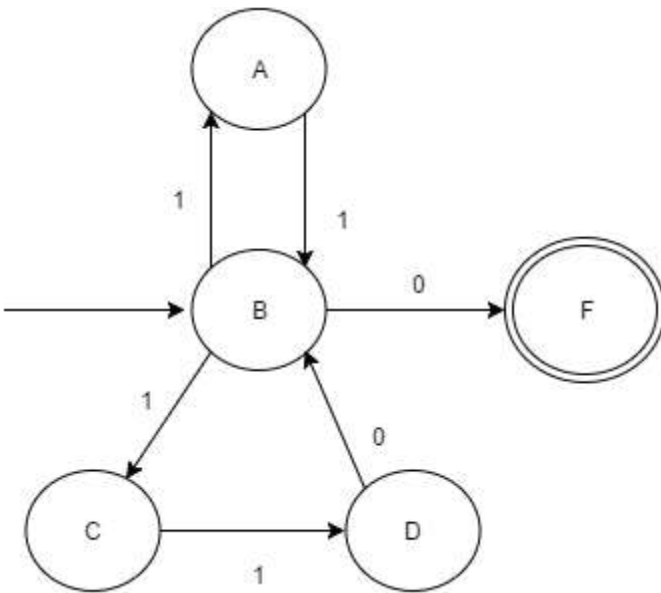


d)



Ans. a

Q16 Which of the following does the given NFA represent?



- a. $\{11, 101\}^* \{01\}$
- b. $\{110, 01\}^* \{11\}$
- c. $\{11, 110\}^* \{0\}$
- d. $\{00, 110\}^* \{1\}$

ans. C

Question 1: While applying Pumping lemma over a language, we consider a string w that belong to L and fragment it into _____ parts.

- a) 2
- b) 5
- c) 3
- d) 6

Ans: c

Question 2: If we select a string w such that $w \in L$, and $w = xyz$. Which of the following portions cannot be an empty string?

- a) x
- b) y
- c) z
- d) all of the mentioned

Ans: b

Question 3: There exists a language L . We define a string w such that $w \in L$ and $w = xyz$ and $|w| \geq n$ for some constant integer n . What can be the maximum length of the substring xy i.e. $|xy| \leq ?$

- a) n
- b) $|y|$
- c) $|x|$
- d) none of the mentioned

Ans: a

Question 4: Fill in the blank in terms of p , where p is the maximum string length in L .

Statement: Finite languages trivially satisfy the pumping lemma by having $n = \underline{\hspace{2cm}}$

- a) $p*1$
- b) $p+1$
- c) $p-1$
- d) None of the mentioned

Ans: b

Question 5: Let w be a string and fragmented by three variable x , y , and z as per pumping lemma. What does these variables represent?

- a) string count
- b) string

- c) both (a) and (b)
- d) none of the mentioned

Ans: a

Question 6: Which of the following is not an application of Pumping Lemma?

- a) $\{0^i1^i \mid i \geq 0\}$
- b) $\{0^ix \mid i \geq 0, x \in \{0, 1\}^* \text{ and } |x| \leq i\}$
- c) $\{0^n \mid n \text{ is prime}\}$
- d) None of the mentioned

Ans: d

Question 7: Pumping lemma method is used to generate from

- a) (i) many input strings (ii) one string
- b) (i) one string (ii) many input strings
- c) Either a or b
- d) None of the mentioned

Ans: a

Question 8: Amongst the various pairs, which pair is taken in the pumping lemma

- a) 1st
- b) 2nd
- c) 4th
- d) None of the mentioned

Ans: a

Question 9: Let $w = xyz$ and y refers to the middle portion and $|y| > 0$. What do we call the process of repeating y 0 or more times before checking that they still belong to the language L or not?

- a) Generating
- b) Pumping
- c) Producing
- d) None of the mentioned

Ans: b

Question 10: Answer in accordance to the third and last statement in pumping lemma:

For all _____ $xyiz \in L$

- a) $i > 0$
- b) $i < 0$
- c) $i \leq 0$
- d) $i \geq 0$

Ans: d

Question 11: Which of the technique can be used to prove that a language is non regular?

- a) Ardens theorem
- b) Pumping Lemma
- c) Ogden's Lemma
- d) None of the mentioned

Ans: b

Question 12: If L_1' and L_2' are regular languages, then $L_1.L_2$ will be

- a) regular
- b) non regular
- c) may be regular

d) none of the mentioned

Ans: a

Question 13: If L_1 and L_2' are regular languages, $L_1 \cap (L_2' \cup L_1)'$ will be

a) regular

b) non regular

c) may be regular

d) none of the mentioned

Ans: a

Question 14: If A and B are regular languages, $!(A' \cup B')$ is:

a) regular

b) non regular

c) may be regular

d) none of the mentioned

Ans: a

Question 15: Which among the following is the closure property of a regular language?

a) Emptiness

b) Universality

c) Membership

d) None of the mentioned

Ans: d

1. There are _____ tuples in finite state machine.

a) 4

b) 5

- c) 6
- d) unlimited

Ans:- B

2. Transition function maps.

- a) $\Sigma * Q \rightarrow \Sigma$
- b) $Q * Q \rightarrow \Sigma$
- c) $\Sigma * \Sigma \rightarrow Q$
- d) $Q * \Sigma \rightarrow Q$

Ans: - d

3. Number of states requires accepting string ends with 10.

- a) 3
- b) 2
- c) 1
- d) can't be represented.

Ans:-a

4. Number of final state requires accepting Φ in minimal finite automata.

- a) 1
- b) 2
- c) 3
- d) None of the mentioned

Ans:-d

5. Number of states requires in NFA simulating a computer with memory capable of storing '3' words each of length '8'.

- a) $3 * 2^8$
- b) $2^{(3*8)}$
- c) $2^{(3+8)}$
- d) None of the mentioned

Ans:- b

6. An NFA's transition function returns

- a. A Boolean value
- b. A state
- c. A set of states
- d. An edge

Ans:C

7. The behaviour of a NFA can be stimulated by DFA

- a. Always
- b. Sometimes
- c. Never
- d. Depends on NFA

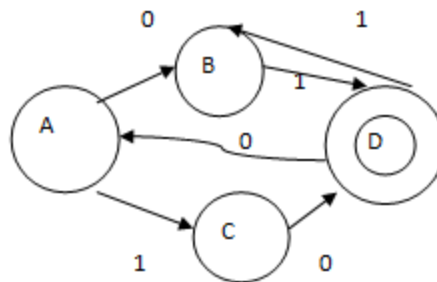
Ans:-a

8. The relation between NFA-accepted languages and DFA accepted languages is

- a. >
- b. <
- c. =
- d. <=

Ans:- c

9. Examine the following DFA: If input is 011100101, which edge is NOT traversed?



- a. A B
- b. C
- c. C D
- d. D A

Ans:- C

10. The transitional function of a NFA is

- a. $Q \times \Sigma \rightarrow Q$

b. $Q \times \Sigma \rightarrow 2Q$

c. $Q \times \Sigma \rightarrow 2n$

d. $Q \times \Sigma \rightarrow Qn$

Ans:- B

11. Which is true for Dead State?

a. It cannot be reached anytime

b. There is no necessity of the state

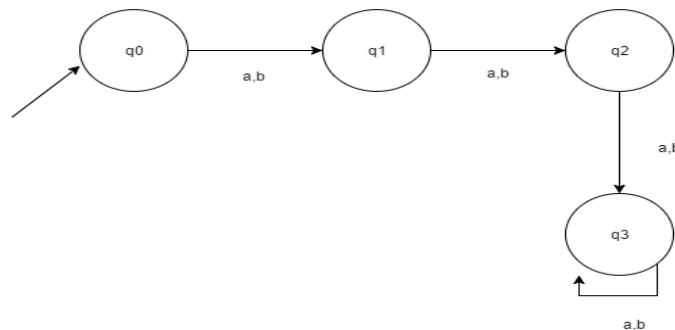
c. If control enters no way to come out from the state

d. If control enters FA deads

Ans:- C

12. Which among the following states would be notated as the final state/acceptance state?

$L = \{x \in \Sigma^* \mid \text{length of } x \text{ is } 2\}$



a. q1

b. q2

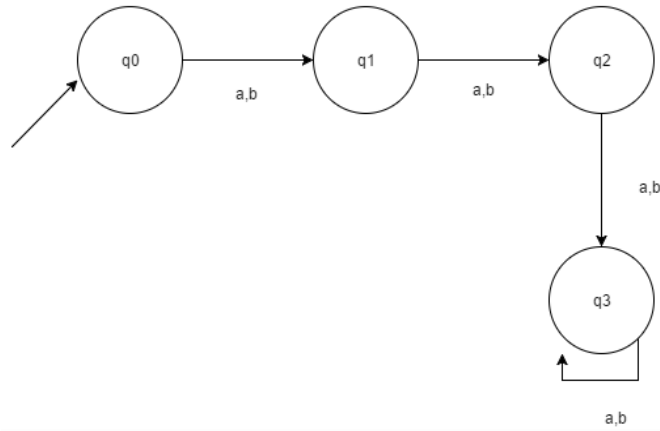
c. q1, q2

d. q3

Ans:- b

13. Which of the following are the final states in the given DFA according to the Language given ?

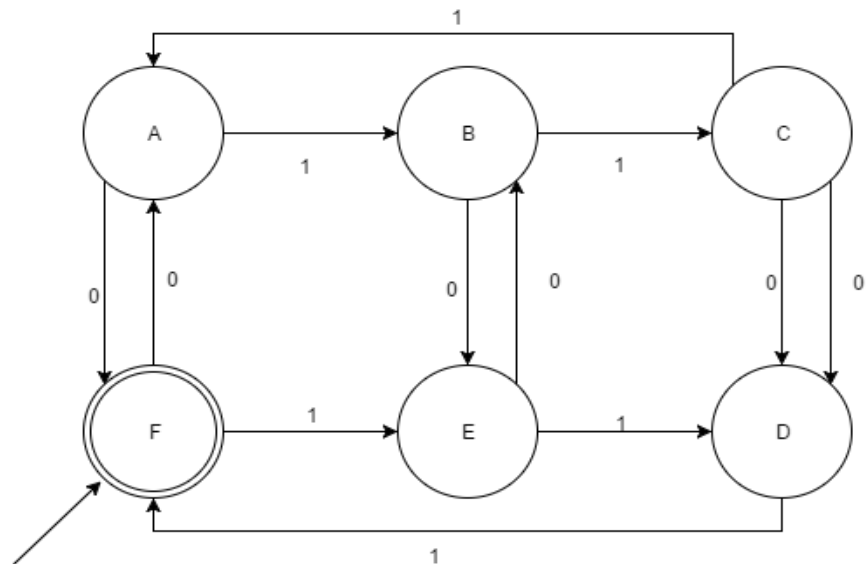
$L = \{x \in \Sigma^* \mid \text{length of } x \text{ is at most } 2\}$



- a. q0, q1
- b. q0, q2
- c. q1, q2
- d. q0, q1, q2

Ans:D

14. Which of the following x is accepted by the given DFA (x is a binary string $\Sigma = \{0,1\}$)?

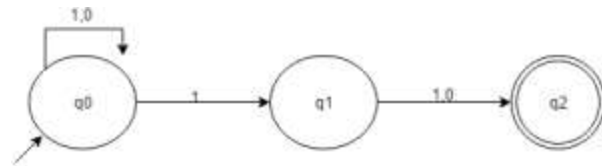


- a. divisible by 3
- b. divisible by 2
- c. divisible by 2 and 3
- d. divisible by 3 and 2

Ans:- d

15. Choose the correct option for the given statement:

Statement: The DFA shown represents all strings which has 1 at second last position.



- a. Correct
- b. Incorrect, Incomplete DFA
- c. Wrong proposition
- d. May be correct

MCQ's on regular languages and regular expressions

1. A language is regular if and only if

- a) **accepted by DFA**
- b) accepted by PDA
- c) accepted by LBA
- d) accepted by Turing machine

2. Regular grammar is

- a) **context free grammar**
- b) non context free grammar
- c) english grammar
- d) none of the mentioned

3. Regular expression are

- a) **Type 0 language**
- b) Type 1 language

c) Type 2 language

d) Type 3 language

4. Consider the languages $L_1 = \phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* \cup L_1^*$

(A) $\{\epsilon\}$

(B) Φ

(C) a^*

(D) $\{\epsilon, a\}$

a) **A**

b) B

c) C

d) D

5. Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ?

1) abaabaaabaa

2) aaaabaaaa

3) baaaaabaaaab

4) baaaaabaa

a) 1, 2 and 3

b) 2, 3 and 4

c) **1, 2 and 4**

d) 1, 3 and 4

6. Let w be any string of length n in $\{0,1\}^*$. Let L be the set of all substrings of w . What is the minimum number of states in a non-deterministic finite automaton that accepts L ?

a) $n-1$

b) n

c) **$n+1$**

d) $2n-1$

7. Which one of the following languages over the alphabet $\{0,1\}$ is described by the regular expression: $(0+1)^*0(0+1)^*0(0+1)^*$?

a) The set of all strings containing the substring 00

b) The set of all strings containing at most two 0's

c) **The set of all strings containing at least two 0's**

d) The set of all strings that begin and end with either 0 or 1.

8. The regular expression $0^*(10^*)^*$ denotes the same set as

a) **$(1^*0)^*1^*$**

b) $0 + (0 + 10)^*$

c) $(0 + 1)^* 10(0 + 1)^*$

d) none of these

9. The smallest finite automation which accepts the language $\{x \mid \text{length of } x \text{ is divisible by } 3\}$ has :

a) 2 states

b) **3 states**

c) 4 states

d) 5 states

10. Consider the following languages

$$L1 = \{ww \mid w \in \{a, b\}^*\}$$

$$L2 = \{ww^R \mid w \in \{a, b\}^*, w^R \text{ is the reverse of } w\}$$

$$L3 = \{0^{2i} \mid i \text{ is an integer}\} \quad L4 = \{0^{i^2} \mid i \text{ is an integer}\}$$

Which of the languages are regular?

a) Only L1 and L2

b) Only L2, L3 and L4

c) Only L3 and L4

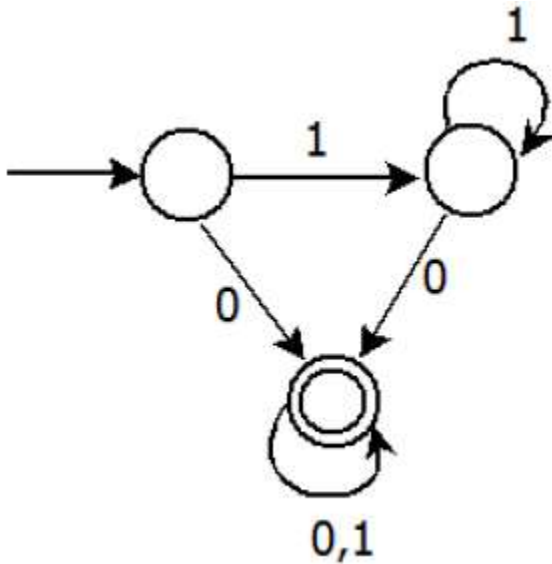
d) **Only L3**

Q1) Consider the languages $L1 = \{a^n \mid n \geq 0\}$ and $L2 = \{a^n \mid n \geq 1\}$. Which one of the following represents $L1 \cap L2$?

a) $\{\epsilon\}$ b) a^* c) $\{\epsilon, a\}$ d) none

Answer: A

Q2)



Consider the DFA given. Which of the following are FALSE?

1. Complement of $L(A)$ is context-free.
2. $L(A) = L((11^*0+0)(0+1)^*0^*1^*)$
3. For the language accepted by A , A is the minimal DFA.
4. A accepts all strings over $\{0, 1\}$ of length at least 2.

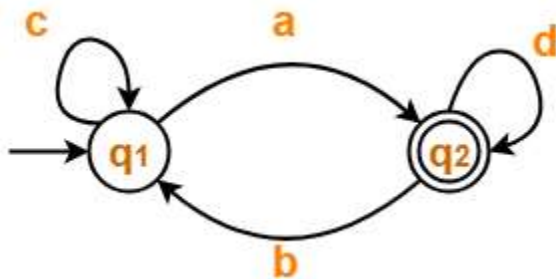
Answer: 3 and 4

Q3) Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ?

- 1) abaabaaabaa
- 2) aaaabaaaa
- 3) baaaaabaaaab
- 4) baaaaabaa

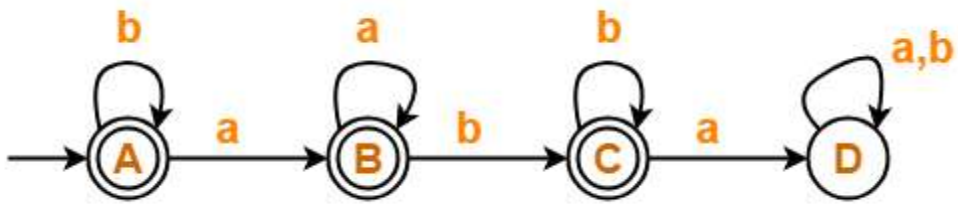
Answer : 1,2,4

Q4) Find the regular expression for the given



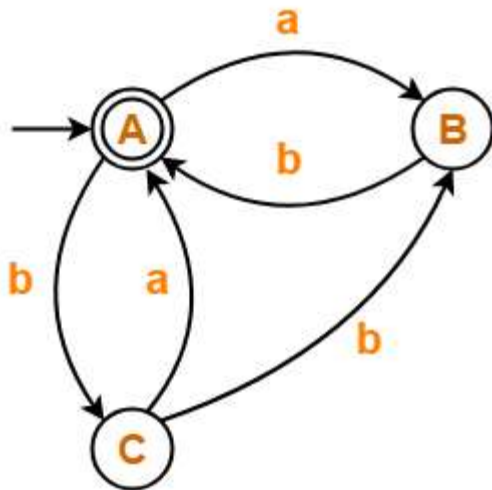
Answer : $c^*a(d+bc^*a)^*$

Q5) Find the regular expression for the given



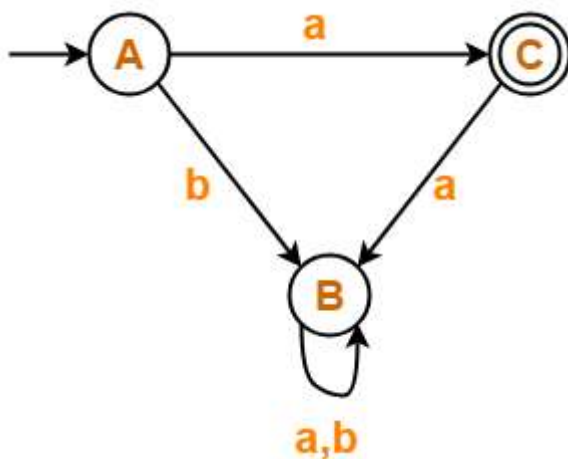
Answer: $b^*(aa^*(bb^*+\epsilon)+\epsilon)$

Q6) Find the regular expression for the given



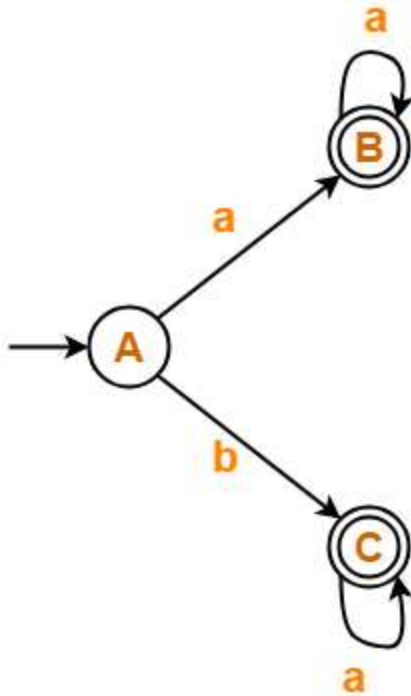
Answer: $(ab + b(a+bb))^*$

Q7) Find the regular expression for the given



Answer: regular expression is only **a**, because state **B** is dead state.

Q8) Find the regular expression for the given



Answer: $aa^* + ba^*$

Q9) A regular language over an alphabet a is one that can be obtained from

- a) union
- b) concatenation
- c) kleene
- d) All of the mentioned

Answer: d

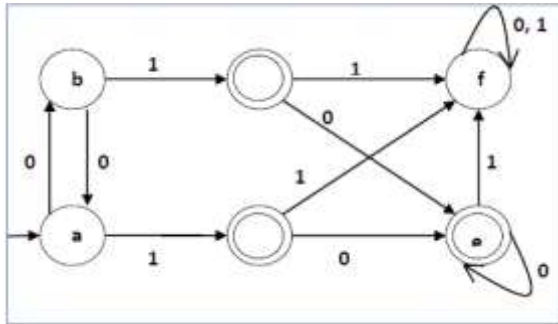
Q10)

Regular expression Φ^* is equivalent to

- a) ϵ
- b) Φ
- c) 0
- d) 1

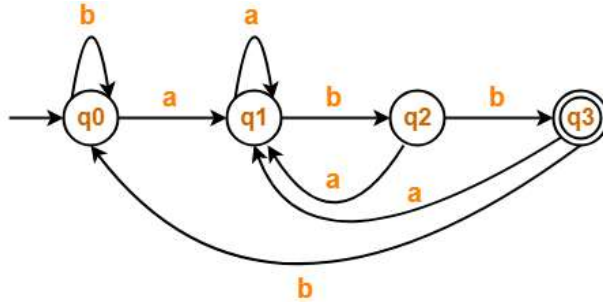
Answer:A

Q: 1 How many states are require to construct a minimal dfa from the given DFA



a)2 b)3 c)4 d)5

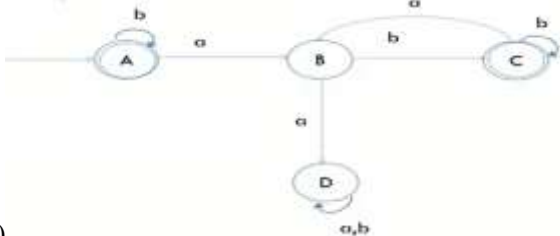
2.From the given DFA which string is accepted



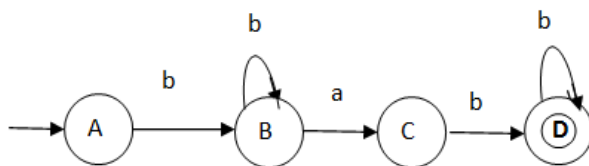
a) abbabb b) abbabba c) aabba d) aaba

ans a

3.Which DFA is accept the string in which every a is followed by b where $\Sigma = \{a,b\}$.



a)



b)

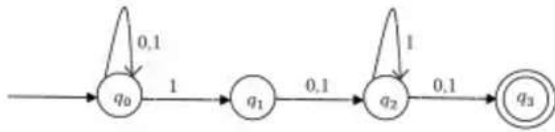


c)

d) None of these

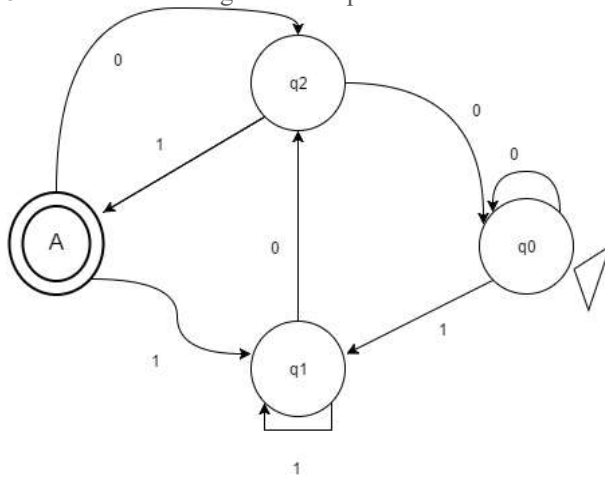
4.

9.) Consider the finite automaton in the following figure. What is the set of reachable states for the input string 0011 ?



- a. q0, q1, q2
- b. q0, q1
- c. q0, q1, q2, q3
- d. q3

5. What the following DFA accepts?



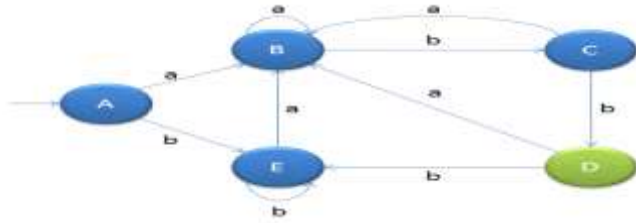
- a) x is a string such that it ends with '101'
- b) x is a string such that it ends with '01'
- c) x is a string such that it has odd 1's and even 0's
- d) x is a strings such that it has starting and ending character as 1

6. Minimize the dfa below given transition table

	a	b
A	B	D
B	C	E
C	B	E
D	C	E
E	E	E

- a) 3 b)4 c)2 d)5

7. How many states are required after construct a minimal DFA



A)3 B)2 C)5 D)4

8)Statement 1: Acceptance of string only when it goes to first state to final state .

statement 2: String is accepted only when it satisfy the language .

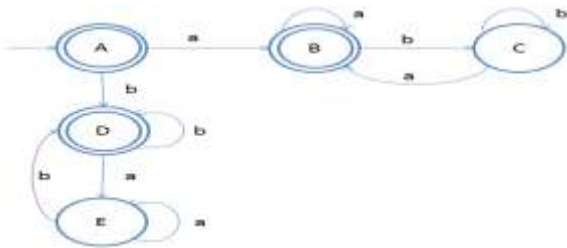
a) Statement 1 is correct , Statement 2 is correct .

b) Statement 1 is not correct , Statement 2 is correct .

c) Statement 1 is correct , Statement is not correct

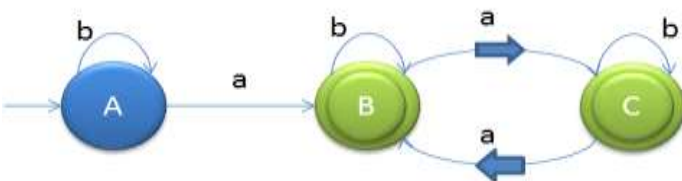
d) Statement 1 is not correct , statement 2 is not correct

9) At which state String 'aa' accepted



a) D b) A c) B d) C

10) How many states after Minimize the below dfa



a) 3 b) 2 c) 4 d) 5

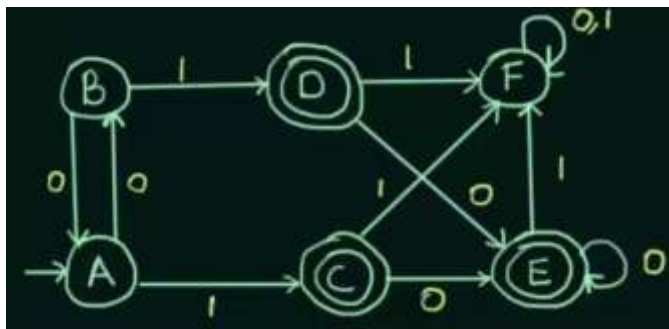
11) Which state is accepted String '110101'

Transition table given below

PS	NS	
	x=0	x=1
→ q ₁	q ₃	q ₂
q ₂	q ₄	q ₁
q ₃	q ₁	q ₄
q ₄	q ₂	q ₃

a) q2 b) q3 c) q1 d) q4

12) Minimize given dfa



a) {A,B}, {F}, {C,D,E}

b) {A,F}, {B}, {C,D,E}

c) {A}, {B,F}, {C,D,E}

d) {A,B,F}, {C,D,E}

13.) Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ?

1) abaabaaabaa

2) aaaabaaaa

3) baaaaabaaaab

4) baaaaabaa

a) 1, 2 and 3

b) 2, 3 and 4

c) 1, 2 and 4

d) 1, 3 and 4

14.) Let w be any string of length n in $\{0,1\}^*$. Let L be the set of all substrings of w . What is the minimum number of states in a non-deterministic finite automaton that accepts L ?

A) $n-1$

b) n

c) $n+1$

D) $2n-1$

15) Which string is accepted by DFA

a) a^n

b) $a^n b^n$

c) $a^n b^n c^n$

1) a, 2) c 3) a,b,c 4) a

1. A language is regular if and only if

a) accepted by DFA

b) accepted by PDA

c) accepted by LBA

d) accepted by Turing machine

Answer- a

2. Regular grammar is

a. context free grammar

b. non context free grammar

c. English grammar

d. none of the mentioned

Answer- a

3. Regular expression are

a. Type 0 language

b. Type 1 language

c. Type 2 language

d. Type 3 language

4. Regular languages are closed under (closure properties)

a. Concatenation

b. Union

c. Complement

d. All of the above

Answer- d

5. If L_1 is regular, then there exists some regular expression r_1 which describes it. Same for L_2 .
Then: $L_1 \cup L_2 = L(r_1) \cup L(r_2) = r_1 + r_2$

$r_1 + r_2$ is a regular expression and therefore describes a regular language.

- a. True
- b. False

Answer : a

6. If $L_1 = \{a^n \mid n \geq 0\}$ and $L_2 = \{b^n \mid n \geq 0\}$ is regular
then $L_3 = L_1.L_2 = \{a^m . b^n \mid m \geq 0 \text{ and } n \geq 0\}$ is also regular?

- a. False
- b True

Answer b

7. Which one of the following languages over the alphabet $\{0,1\}$ is described by the regular expression?

$(0+1)^*0(0+1)^*0(0+1)^*$

- a. The set of all strings containing the substring 00.
- b. The set of all strings containing at most two 0's.
- c. The set of all strings containing at least two 0's.
- d. The set of all strings that begin and end with either 0 or 1.

Answer c

8. A language is regular if it can be expressed in terms of

- a. Regular expression
- b. Type 0 Language
- c. English Grammar
- g. None of the above

Answer a

9. Which of the following pairs of regular expressions are equivalent?

- a. $1(01)^*$ and $(10)^*1$
- b. $x(xx)^*$ and $(xx)^*x$
- c. x^+ and $x^+x^+(^+)$
- d. All of the above mentioned

Answer d

10. Regular expressions are used to represent which language

- a) Recursive language
- b) Context free language
- c) Regular language
- d) All of these

Answer c

11. Which of the following operation can be applied on regular expressions?

- a) Union
- b) Concatenation
- c) Closure
- d) All of these

answer d

12. 7. Which of the following identity is wrong?

- a) $R + R = R$
- b) $(R^*)^* = R^*$
- c) $\epsilon R = R\epsilon = R$
- d) $\emptyset R = R\emptyset = RR^*$

Answer d

13. Which of the following statement is true?

- a) Every language that is defined by regular expression can also be defined by finite automata
- b) Every language defined by finite automata can also be defined by regular expression
- c) We can convert regular expressions into finite automata
- d) All of these

Answer d

14. Which of the following identity is true?

- a) $\epsilon + RR^* = R^* = \epsilon + R^*R$
- b) $(R_1R_2)^*R_1 = R_1(R_2R_1)^*$

c) $R^*R^* = R^*$

d) All of these

15. $(a+b)^*$ is equivalent to

a) b^*a^*

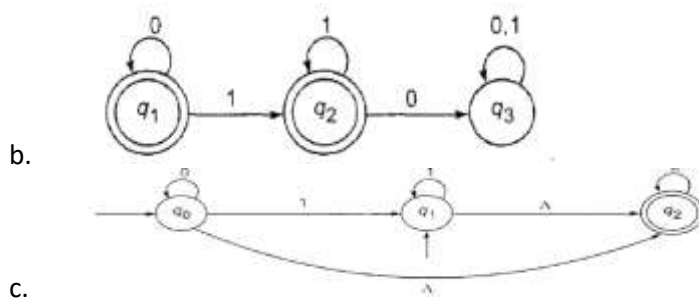
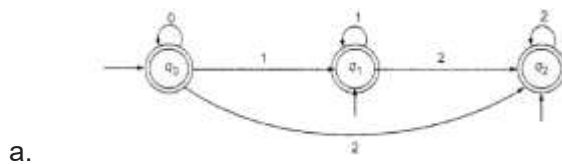
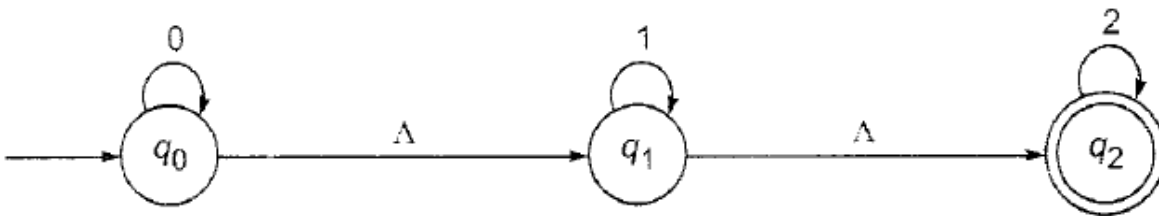
b) $(a^*b^*)^*$

c) a^*b^*

d) none of the mentioned

Answer b

16. Consider a finite automaton, with A-moves, given in Fig. which among the following is the Equivalent automaton without A-moves.



d. None of above

Answer a

17. Suppose we want to replace a null move from vertex V1 to vertex V2 in FA Then

What is the first step needs to be done:

a. Make a new vertex V3

b. Find all the edges starting from v2

- c. Delete Vertex V2
- d. None of the above

Answer b

A context sensitive language is accepted by

- a) Finite automata
- b) Linear bounded automata
- c) Both (a) and (b)
- d) None of these

Ans- B

A grammar $G = (V, T, P, S)$ in which V is

- a) Set of variables
- b) Set of terminals
- c) Set of variables and terminals
- d) None of these

Ans-A

A grammar $G = (V, T, P, S)$ in which T is

- a) Set of variables
- b) Set of terminals
- c) Set of variables and terminals
- d) None of these

Ans- B

Which of the following is more powerful?

- a) PDA
- b) Turing machine
- c) Finite automata
- d) Context sensitive language

Ans-B

Which of the following string is generated by the grammar

$S \rightarrow 0S1AB$

$A \rightarrow 4A$

$B \rightarrow SA$

- a) 0111
- b) 010101
- c) 00000

d) None of these

Which of the following relates to Chomsky hierarchy?

- a) Regular < CFL < CSL < Unrestricted
- b) CFL < CSL < Unrestricted < Regular
- c) CSL < Unrestricted < CF < Regular
- d) None of the mentioned

Ans-a

A language is accepted by a push down automata if it is:

- a) regular
- b) context free
- c) both (a) and (b)
- d) none of the mentioned

Ans- B

Which of the following strings do not belong the given regular expression?

(a)*(a+cba)

- a) aa
- b) aaa
- c) acba
- d) acbacba

Ans-d

Which of the following CFG's can't be simulated by an FSM ?

A. $S \rightarrow Sa \mid b$

B. $S \rightarrow aSb \mid ab$

C. $S \rightarrow abX, X \rightarrow cY, Y \rightarrow d \mid aX$

D. None of these

Ans-B

Regular Grammar is accepted by

- a. FA
- b. LBA
- c. Turing Machine

d. All

Q. If $\Sigma = \{a,b\}$ and given productions are

$$S \rightarrow XaaX$$

$$X \rightarrow aX \mid bX \mid \Lambda$$

Then the above grammar defines the language expressed by _____ regular expression

- e. $(a+b)^*aa(a+b)^*$
- f. $(a+b)^*a(a+b)^*a$
- g. $(a+b)^*aa(a+b)^*aa$
- h. $(a+b)^*aba+b)^*$

Ans-A

Q. The languages which can not accept PDA _____

- A. REGULAR LANGUAGES
- B. CONTEXT FREE
- C. CONTEXT SENSITIVE
- D. NONE.

Ans- C

Q Regular grammar is

- a) Context free grammar
- b) Non context free grammar
- c) English grammar
- d) None of the mentioned

Ans- A

Regular expression are

- a) Type 0 language
- b) Type 1 language
- c) Type 2 language
- d) Type 3 language

Ans- A

L and $\sim L$ are recursive enumerable then L is

- a) Regular
- b) Context free
- c) Context sensitive
- d) Recursive

Ans-D

Regular expressions are closed under

- a) Union
- b) Intersection
- c) Kleene star

d) All of the mentioned

Ans-d

How many strings of length less than 4 contains the language described by the regular expression $(x+y)^*y(a+ab)^*$?

- a) 7
- b) 10
- c) 12
- d) 11

Ans-D

Grammatical rules which involve the meaning of words are called:

- a. Semantics
- b. Syntactic
- c. Both a and b
- d. None of given

Ans- A

Grammatical rules which do not involve the meaning of words are called:

- a. Semantics
- b. Syntactic
- c. Both a and b
- d. None of given

Ans-B

The symbols in a grammar that can't be replaced by anything are called:

- a. Productions
- b. Terminals
- c. Non-terminals
- d. All of above

Ans-B

The symbols in a grammar that must be replaced by other Symbols are called:

- e. Productions
- f. Terminals
- g. Non-terminals
- h. None of given

Ans-C

Q1. Consider the languages $L_1 = \emptyset$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* \cup L_1$

a) $\{\epsilon\}$ b) $\emptyset$ c) a^* d) $\{\epsilon, a\}$

Q2. Let S and T be language over $\{a, b\}$ represented by the regular expressions $(a+b)^*$ and $(a+b)^*$, respectively. Which of the following is true?

(a) $S \subset T$ (S is a subset of T)

(b) $T \subset S$ (T is a subset of S)

(c) $S=T$

(d) $S \cap T = \emptyset$

Q3. Let L denotes the language generated by the grammar $S \rightarrow OSO/00$. Which of the following is true?

(a) $L = O$

(b) L is regular but not $\emptyset$

(c) L is context free but not regular

(d) L is not context free

Q4. Consider the following two statements:

S1: $\{0^{2n} \mid n \geq 1\}$ is a regular language

S2: $\{0^m 0^n 0^{(m+n)} \mid m \geq 1 \text{ and } n \geq 2\}$ is a regular language

Which of the following statements is correct? (GATE CS 2001)

a) Only S1 is correct

b) Only S2 is correct

c) Both S1 and S2 are correct

d) None of S1 and S2 is correct

Q5. Which of the following statements in true? (GATE CS 2001)

(a) If a language is context free it can always be accepted by a deterministic push-down automaton

(b) The union of two context free languages is context free

(c) The intersection of two context free languages is context free

(d) The complement of a context free language is context free

Q6. Given an arbitrary non-deterministic finite automaton (NFA) with N states, the maximum number of states in an equivalent minimized DFA is at least. (GATE CS 2001)

(a) N^2

(b) 2^N

(c) $2N$

(d) $N!$

Q7. Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ?

1) abaabaaabaa

2) aaaabaaaa

3) baaaaabaaaab

4) baaaaabaa

(A) 1, 2 and 3

(B) 2, 3 and 4

(C) 1, 2 and 4

(D) 1, 3 and 4

8. Which of the following problems are decidable?

....1) Does a given program ever produce an output?

....2) If L is a context-free language, then is L' (complement of L) also context-free?

....3) If L is a regular language, then is L' also regular?

....4) If L is a recursive language, then, is L' also recursive?

(A) 1, 2, 3, 4

(B) 1, 2,

(C) 2, 3, 4

(D) 3, 4

9. Let P be a regular language and Q be context-free language such that $Q \subseteq P$. (For example, let P be the language represented by the regular expression p^*q^* and Q be $\{pnqn\}$

$|n \in \mathbb{N}\rangle$). Then which of the following is ALWAYS regular?

(A) $P \cap Q$

(B) $P - Q$

(C) $\Sigma^* - P$

(D) $\Sigma^* - Q$

10. $(a+b)^*$ is equivalent to

a) b^*a^*

b) $(a^*b^*)^*$

c) a^*b^*

d) none of the mentioned

11. $a^?$ is equivalent to

a) a

b) $a+\Phi$

c) $a+\epsilon$

d) wrong expression

12. Regular expression Φ^* is equivalent to

a) ϵ

b) Φ

c) 0

d) 1

13. A language is regular if and only if

a) accepted by DFA

b) accepted by PDA

c) accepted by LBA

d) accepted by Turing machine

14. Let the class of language accepted by finite state machine be L_1 and the class of languages represented by regular expressions be L_2 then

a) $L_1 < L_2$

b) $L1 \supseteq L2$

c) $L1 \cup L2 = .^*$

d) $L1 = L2$

15. Regular expression $\{0,1\}$ is equivalent to

a) $0 \cup 1$

b) $0 / 1$

c) $0 + 1$

d) All of the mentioned

1. Transition function of DFA maps.

a) $\Sigma * Q \rightarrow \Sigma$

b) $Q * Q \rightarrow \Sigma$

c) $\Sigma * \Sigma \rightarrow Q$

d) $Q * \Sigma \rightarrow Q$

Answer: d

2. Minimum Number of states require to accept string ends with 10.

a) 3

b) 2

c) 1

d) can't be represented.

Answer: a

3. $\delta^*(q, ya)$ is equivalent to .

a) $\delta((q, y), a)$

b) $\delta(\delta^*(q, y), a)$

c) $\delta(q, ya)$

d) independent from δ notation

Answer: b

4. Languages of a automata is

a) If it is accepted by automata

b) If it halts

- c) If automata touch final state in its life time
- d) All language are language of automata

Answer: a

5. Finite automata requires minimum _____ number of stacks.
- a) 1
 - b) 0
 - c) 2
 - d) None of the mentioned

Answer: b

6. FSM with output capability can be used to add two given integer in binary representation. This is
- a) True
 - b) False
 - c) May be true
 - d) None of the mentioned

Answer: a

7. If NFA of 6 states excluding the initial state is converted into DFA, maximum possible number of states for the DFA is ?
- a) 64
 - b) 32
 - c) 128
 - d) 127

Answer: c

8. NFA, in its name has 'non-deterministic' because of :
- a) The result is undetermined
 - b) The choice of path is non-deterministic
 - c) The state to be transited next is non-deterministic
 - d) All of the mentioned

Answer: b

9. The construction time for DFA from an equivalent NFA (m number of node) is:
- a) $O(m^2)$
 - b) $O(2^m)$
 - c) $O(m)$
 - d) $O(\log m)$

Answer: b

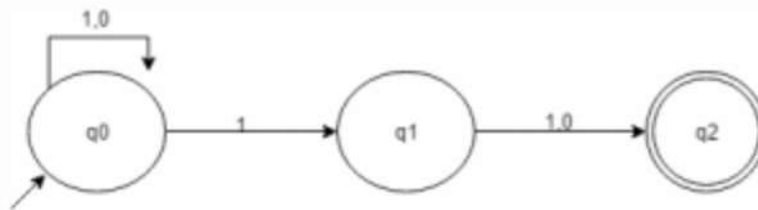
10. Which of the following option is correct?

- a) NFA is slower to process and its representation uses more memory than DFA
- b) DFA is faster to process and its representation uses less memory than NFA
- c) NFA is slower to process and its representation uses less memory than DFA
- d) DFA is slower to process and its representation uses less memory than NFA

Answer: c

11. Choose the correct option for the given statement:

Statement: The DFA shown represents all strings which has 1 at second last position.



- a) Correct
- b) Incorrect, Incomplete DFA
- c) Wrong proposition
- d) May be correct

Answer: c

12.

TOC mcq

Definition and description of FA:

1. NFA, in its name has 'non-deterministic' because of :

- a) The result is undetermined
- b) The choice of path is non-deterministic
- c) The state to be transited next is non-deterministic
- d) All of the mentioned

Answer: b

2. Languages of a automata is

- a) If a string is accepted by automata
- b) If it halts
- c) If automata touch final state in its life time

d) All language are language of automata

Answer: a

3. The basic limitation of finite automata is that

a) It can't remember arbitrary large amount of information.

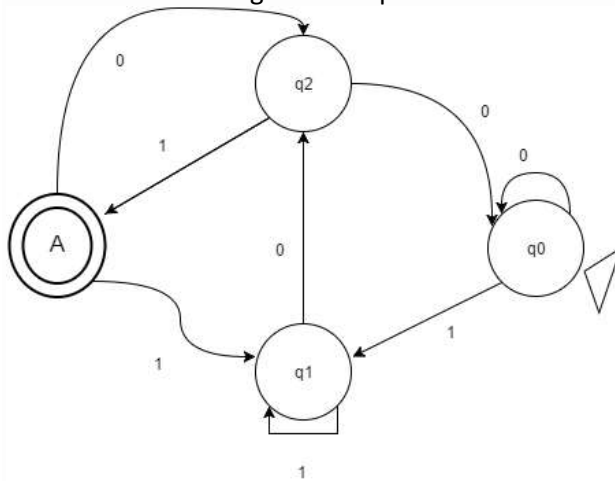
b) It sometimes recognize grammar that are not regular.

c) It sometimes fails to recognize regular grammar.

d) All of the mentioned

Answer:a

4. What the following DFA accepts?



a) x is a string such that it ends with '101'

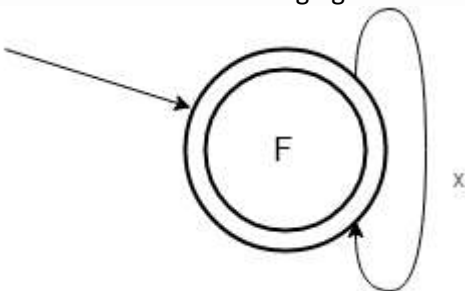
b) x is a string such that it ends with '01'

c) x is a string such that it has odd 1's and even 0's

d) x is a strings such that it has starting and ending character as 1

Answer A)

5. What does the following figure most correctly represents?



a) Final state with loop x

b) Transitional state with loop x

c) Initial state as well as final state with loop x

d) Insufficient Data

Answer: c

6. Number of final state requires to accept Φ in minimal finite automata.

- a) 1
- b) 2
- c) 3
- d) None of the mentioned

Answer:d

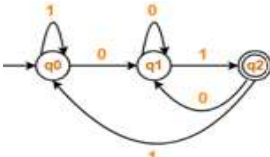
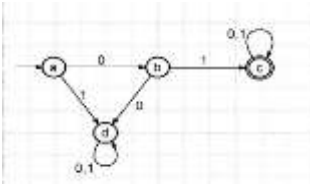
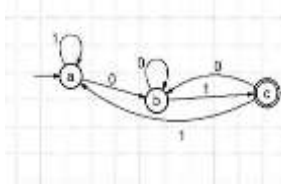
Explanation: No final state requires.

7. Which of the following is an application of Finite Automaton?

- a) Compiler Design
- b) Grammar Parsers
- c) Text Search
- d) All of the mentioned

Answer: d

8. Which DFA is correct for the given Language: accepting string ending with '01' over input alphabets $\Sigma = \{a, b\}$

- a) 
- b) 
- c) 
- d) None of these Ans c

9. How many Final states in the given language : $\{a^n \mid \text{where } n \geq 1 \text{ but } n \neq 2, 6\}$, $\Sigma = \{a\}$

- a) 2
- b) 4
- c) 3
- d) 6

Ans b)

10. Transition function maps.

- a) $\Sigma * Q \rightarrow \Sigma$
- b) $Q * Q \rightarrow \Sigma$
- c) $\Sigma * \Sigma \rightarrow Q$
- d) $Q * \Sigma \rightarrow Q$

Ans d)

11. Finite automata has only

- a) Finite memory
- b) Read only head
- c) Finite control
- d) All of them

Ans d)

The Equivalence of deterministic and non-deterministic finite automata

12. The relation between NFA-accepted languages and DFA accepted languages is

- a) $>$
- b) $<$
- c) $=$
- d) $\leq$

Ans c)

13. Which of the following is correct proposition?

Statement 1: Non determinism is a generalization of Determinism.

Statement 2: Every DFA is automatically an NFA

a) Statement 1 is correct because Statement 2 is correct

b) Statement 2 is correct because Statement 2 is correct

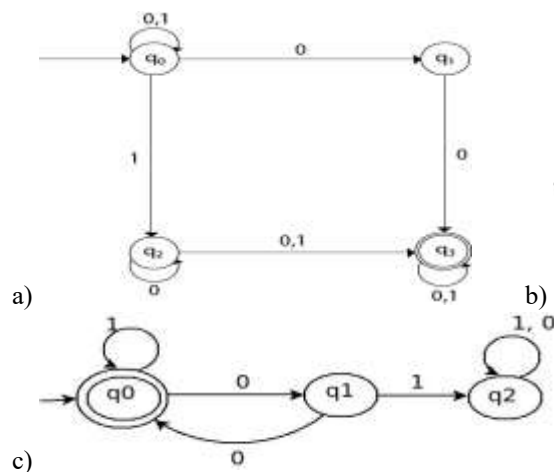
c) Statement 2 is false and Statement 1 is false

d) Statement 1 is false because Statement 2 is false

Ans a)

14. Which NFA is correct for the transition table as given below:

Present State	0	1
$\rightarrow q_0$	q_0, q_1	q_0, q_2
q_1	q_3	ϵ
q_2	q_2, q_3	q_3
$\rightarrow q_3$	q_3	q_3



Ans a)

15. Given a NFA with N states, the maximum number of states in an equivalent minimized DFA is at least

a) N^2

b) 2^N

c) $2N$

d) $N!$

Ans b)

Q1 What is wrong in the given definition?

Def: $(\{q_0, q_1, q_2\}, \{0,1\}, \delta, q_3, \{q_3\})$

a) The definition does not satisfy 5 Tuple definition of NFA

- b) There are no transition definition
- c) Initial and Final states do not belong to the Graph
- d) Initial and final states can't be same

Answer: c

Q2) Number of states require to accept string ends with 10.

- a) 3
- b) 2
- c) 1
- d) can't be represented.

Answer: a

Q3) $\delta(q, ya)$ is equivalent to .

- a) $\delta((q, y), a)$
- b) $\delta(\delta(q, y), a)$
- c) $\delta(q, ya)$
- d) independent from δ notation

answer: b

Q4 Finite automata requires minimum _____ number of stacks.

- a) 1
- b) 0
- c) 2

d) None of the mentioned

answer: b

Q 5 An NFA's transition function returns

a. A Boolean value

b. A state

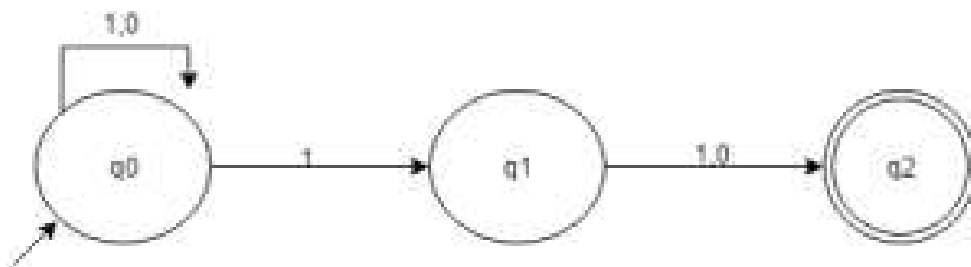
c. A set of states

d. An edge

answer: c

Q6 Choose the correct option for the given statement:

Statement: The DFA shown represents all strings which has 1 at second last position.



a) Correct

b) Incorrect, Incomplete DFA

c) Wrong proposition

d) May be correct

Answer: c (bcz it is nfa)

Q7. Which of the following is a not a part of 5-tuple finite automata?

a) Input alphabet

b) Transition function

c) Initial State

d) Output Alphabet

Answer: d

Q8) Transition function maps.

a) $\Sigma * Q \rightarrow \Sigma$

b) $Q * Q \rightarrow \Sigma$

c) $\Sigma * \Sigma \rightarrow Q$

d) $Q * \Sigma \rightarrow Q$

Answer: d

Q9 The sum of minimum and maximum number of final states for a DFA having 'n' states is equal to:

a) $n+1$

b) n

c) $n-2$

d) $n+2$

answer : a

Q 10 The maximum number of transitions which can be performed over a state in a DFA using $\Sigma = \{a, b, c\}$?

a) 1

b) 2

c) 3

d) 4

answer : c

5. Given an arbitrary non-deterministic finite automaton (NFA) with N states, the maximum number of states in an equivalent minimized DFA is at least. (GATE CS 2001)

- (a) N^2
- (b) 2^N
- (c) $2N$
- (d) $N!$

Answer: (b)

References:

http://en.wikipedia.org/wiki/Powerset_construction

Consider $L = \{(a^p)^* \mid p \text{ is a prime number}\}$ over the alphabet $\{a\}$, then what is the minimum number of states in NFA that accepts the language L?

- (A) three
- (B) five
- (C) four
- (D) six

Explanation:

$L = \{(a^p)^* \mid p \text{ is a prime number}\}$

$L = (a^2)^* \cup (a^3)^* \cup (a^5)^* \cup \dots$

$L = \{\text{epsilon}, a^2, a^3, a^4, a^5, \dots\}$

That is, all strings of a's except 'a'

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7:53 AM 8/12/2019

In the above NFA, only string 'a' is not accepted. Hence, number of states required is three.

So, option (A) is correct.

Let w be any string of length n in $\{a, b\}^*$. Consider 'L' be the set of all strings ending with at least n a's. What is the minimum number of states in non deterministic finite automata that accept 'L'?

- (A) $(n+3)$
- (B) $(n+1)$
- (C) n
- (D) 2^n

Explanation:

It is correct since, the minimum number of states required for NFA for ending with at least 2 a's is $(2 + 1)$ i.e., regular expression will be $(a + b)^*aa$

Hence, Number of states required for at least n a's will be $(n+1)$.

Option (B) is true.

What is the minimum number of states in deterministic finite automata (DFA) for string starting with ba^2 and ending with 'a' over alphabet $\{a, b\}$?

- (A) Ten
- (B) Nine
- (C) Eight
- (D) Six

Option (B) is true.

Ques-3: What is the minimum number of states in deterministic finite automata (DFA) for string starting with ba^2 and ending with 'a' over alphabet $\{a, b\}$?

(A) Ten
(B) Nine
(C) Eight
(D) Six

Explanation:

```

graph LR
    0((0)) -- b --> 1((1))
    1 -- a --> 2((2))
    2 -- a --> 3((3))
    3 -- b --> 4((4))
    4 -- a --> 5((5))
    5 -- b --> 4
    style 5 fill:#00ff00
  
```

In the above DFA, minimum number of states required is six.

n the above DFA, minimum number of states required is six.

Option (D) is correct.

Ques-5: What is the number of states in minimal NFA(non deterministic finite automata), which accepts set of all strings in which the third last symbol is 'a' over alphabet $\{a, b\}$?

- (A) three
- (B) four
- (C) six
- (D) five

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(D) none

Explanation:
Both given languages are regular. Option (C) is correct.

Ques-5: What is the number of states in minimal NFA(non deterministic finite automata), which accepts set of all strings in which the third last symbol is 'a' over alphabet {a, b}?

(A) three
(B) four
(C) six
(D) five

Explanation:

In the above NFA, minimum number of states required is four.
Option (B) is true.

Second Normal Form (2NF)
First Normal Form (1NF)
Third Normal Form (3NF)

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3) Definition of a language L with alphabet $\{a\}$ is given as following. $L = \{a^k \mid k > 0, \text{ and } n \text{ is a positive integer constant}\}$ What is the minimum number of states needed in a DFA to recognize L ?

- (A) $k+1$
- (B) $n+1$
- (C) 2^{n+1}
- (D) 2^{k+1}

Answer (B)

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Differences between POP3 and IMAP
Compiler Design | Recursive Descent Parser
Compiler Design | Loop Optimization

(B) $n+1$
(C) 2^{n+1}
(D) 2^{k+1}

Answer (B)
Note that n is a constant and k is any positive integer. For example, if n is given as 3, then the DFA must be able to accept $3a, 6a, 9a, 12a, \dots$. To build such a DFA, we need 4 states.

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- A) $k+1$
- (B) $n+1$
- (C) 2^{n+1}
- (D) 2^{k+1}

Answer (B)

Note that n is a constant and k is any positive integer. For example, if n is given as 3, then the DFA must be able to accept $3a, 6a, 9a, 12a, \dots$. To build such a DFA, we need 4 states.

4) Let w be any string of length n is $\{0,1\}^*$. Let L be the set of all substrings of w . What is the minimum number of states in a non-deterministic finite automaton that accepts L ?

- (A) $n-1$
- (B) n
- (C) $n+1$
- (D) $2n-1$

Answer (C)

We need minimum $n+1$ states to build NFA that accepts all substrings of a binary string. For example, following NFA accepts all substrings of "010" and it has 4 states.

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(D) 2n-1

Answer (C)

We need minimum $n+1$ states to build NFA that accepts all substrings of a binary string. For example, following NFA accepts all substrings of "010" and it has 4 states.

```

graph LR
    start(( )) --> q0((q0))
    q0 -- 0 --> q1((q1))
    q1 -- 1 --> q2((q2))
    q2 -- 0 --> q3((q3))
    q1 -- 1 --> q1
    style start fill:none,stroke:none
  
```

NFA that accepts all substrings of 010
The substrings are: 0, 1, 01, 10, 010

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8:09 AM 6/12/2019

1) The lexical analysis for a modern language such as Java needs the power of which one of the following machine models in a necessary and sufficient sense?

- (A) Finite state automata
- (B) Deterministic pushdown automata
- (C) Non-deterministic pushdown automata
- (D) Turing machine

Answer (A)

Lexical analysis is the first step in compilation. In lexical analysis, program is divided into tokens. Lexical analyzers are typically based on finite state automata. Tokens can typically be expressed as different regular expressions:

An identifier is given by $[a-zA-Z][a-zA-Z0-9]^*$

The keyword if is given by if.

Integers are given by $[+ -]?[0-9]^+$.

3) A deterministic finite automation (DFA) D with alphabet $\Sigma = \{a, b\}$ is given below

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Computer Networks | Set 14

Last Minute Notes - Discrete Mathematics

Cache Memory Design

Difference between Schema and Database

Difference between Super Key and Candidate Key

Difference between Normalization and Denormalization

Operating System | Recovery From Deadlock

Differences between POP3 and IMAP

3) A deterministic finite automation (DFA) D with alphabet $\Sigma = \{a, b\}$ is given below

Which of the following finite state machines is a valid minimal DFA which accepts the same language as D ?

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Which of the following finite state machines is a valid minimal DFA which accepts the same language as D ?

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Denormalization

Operating System | Recovery From Deadlock

Differences between POP3 and IMAP

Compiler Design | Recursive Descent Parser

Compiler Design | Loop Optimization

Difference between Multiprocessing and Multithreading

Devices used in each layer of TCP/IP model

Collision-Free Protocols in Computer Network

(A)

(B)

(C)

(D)

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Third Normal Form (3NF)

Last Minute Notes Computer Organization

Weak Entity Set in ER diagrams

Difference between Strong and Weak Entity

Difference between Long-Term and Short-Term Scheduler

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Answer (A)

Options (B) and (C) are invalid because they both accept 'b' as a string which is not accepted by given DFA. D is invalid because it accepts $bb+a$ which are not accepted by given DFA.

Following questions have been asked in GATE CS 2012 exam.

1) What is the complement of the language accepted by the NFA shown below? Assume $\Sigma = \{a\}$ and ϵ is the empty string

The screenshot shows a web browser window with the URL <https://www.geekforgeeks.org/automata-theory-qa-2/>. The page displays a question about the complement of a language accepted by an NFA. The NFA diagram shows a start state with a self-loop labeled 'a' and a transition to an accepting state labeled 'ε'. The question asks for the complement of the language accepted by the NFA, given $\Sigma = \{a\}$ and ϵ is the empty string. The options are (A) Φ , (B) ϵ , (C) a , and (D) $\{a, \epsilon\}$. The answer is (B) ϵ . The explanation states that the given alphabet Σ contains only one symbol $\{a\}$ and the given NFA accepts all strings with any number of occurrences of 'a'. In other words, the NFA accepts a^+ . Therefore, the complement of the language accepted by automata is empty string.

2) Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ?

...1) $abaaabaaabaa$
...2) $aaaaabaaaa$
...3) $baaaaaabaaaa$

(A) Φ

(B) ϵ

(C) a

(D) $\{a, \epsilon\}$

Answer (B)

The given alphabet Σ contains only one symbol $\{a\}$ and the given NFA accepts all strings with any number of occurrences of 'a'. In other words, the NFA accepts a^+ . Therefore, the complement of the language accepted by automata is empty string.

4) Consider the set of strings on $\{0,1\}$ in which, every substring of 3 symbols has at most two zeros. For examples, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed DFA that accepts this language is shown below.

The missing arcs in the DFA are

(A)

	00	01	10	11	q
00	0	1			
01				1	
10					
11			0		

(B)

	00	01	10	11	q
00	0				
01		1			
10				0	
11			0		

(C)

	00	01	10	11	q
00	0				
01		1			
10				0	
11			0		

(D)

	00	01	10	11	q
00	0				
01		1			
10				0	
11			0		

The missing arcs in the DFA are

The missing arcs in the DFA are

(A)

	00	01	10	11	q
00	0	1			
01				1	
10					
11			0		

(B)

	00	01	10	11	q
00	0				
01		1			
10				0	
11			0		

(C)

	00	01	10	11	q
00	0				
01		1			
10				0	
11			0		

(D)

	00	01	10	11	q
00	0				
01		1			
10				0	
11			0		

Answer (D)
State 'q' is trap state. All other states are accept states. In state 00, DFA must move to 'q' for input symbol 0. All (non-trap) states indicate names indicate the characters seen before reaching that particular state. Option (D) is the only option that follow these rules.

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The missing arcs in the DFA are

(A)

	00	01	10	11	q
00	1	0			
01			1		
10				1	
11				0	

(B)

	00	01	10	11	q
00		0			
01	1				1
10				0	
11	0				

(C)

	00	01	10	11	q
00		1			0
01			1		
10				1	
11	0				

(D)

	00	01	10	11	q
00	1				
01		1			0
10			1		
11				1	

Answer (D)
State 'q' is trap state. All other states are accept states. In state 00, DFA must move to 'q' for input symbol 0. All (non-trap) states indicate names indicate the characters seen before reaching that particular state. Option (D) is the only option that follow these rules.
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Answer (D)

State 'q' is trap state. All other states are accept states. In state 00, DFA must move to 'q' for input symbol 0. All (non-trap) states indicate names indicate the characters seen before reaching that particular state. Option (D) is the only option that follows these rules.

Ques-1: Consider the following statements:

- X: For any language either a language L or its compliment L' must be finite.
- Y: DFA for language which contains epsilon must have initial state as final state.
- Z: Non-deterministic finite automata is more powerful than deterministic finite automata.

<https://www.sanfoundry.com/automata-theory-mcqs-finite-automata/>

2. In DFA Transition function maps.

- $\Sigma * Q \rightarrow \Sigma$
- $Q * Q \rightarrow \Sigma$
- $\Sigma * \Sigma \rightarrow Q$
- $Q * \Sigma \rightarrow Q$

Answer:d

3. Number of states require to accept string ends with 10.

- 3
- 4

- c) 5
- d) can't be represented.

Answer :a

8. Language of finite automata is.

- a) Type 0
- b) Type 1
- c) Type 2
- d) Type 3

View Answer

Answer:d

Explanation: According to Chomsky classification.

Finite automata requires minimum _____ number of stacks.

- a) 1
- b) 0
- c) 2
- d) None of the mentioned

View Answer

Answer:b

Explanation: Finite automata doesn't require any stack operation

13. The basic limitation of finite automata is that

- a) It can't remember arbitrary large amount of information.
- b) It sometimes recognize grammar that are not regular.
- c) It sometimes fails to recognize regular grammar.
- d) All of the mentioned

Answer:a

Explanation:Because there is no memory associated with automata.

15. FSM with output capability can be used to add two given integer in binary representation.

This is

- a) True
- b) False
- c) May be true
- d) None of the mentioned

View Answer

Answer:a

Explanation: Use them as a flip flop output

5. NFA, in its name has 'non-deterministic' because of :

- a) The result is undetermined

- b) The choice of path is non-deterministic
- c) The state to be transited next is non-deterministic
- d) All of the mentioned

View Answer

Answer: b

Explanation: Non deterministic or deterministic depends upon the definite path defined for the transition from one state to another or undefined(multiple paths).

1. Which of the following options is correct?

Statement 1: Initial State of NFA is Initial State of DFA.

Statement 2: The final state of DFA will be every combination of final state of NFA.

- a) Statement 1 is true and Statement 2 is true
- b) Statement 1 is true and Statement 2 is false
- c) Statement 1 can be true and Statement 2 is true
- d) Statement 1 is false and Statement 2 is also false

View Answer

Answer: a

Explanation: Statement 1 and 2 always true for a given Language.

6. Convert the NFA to DFA

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Info with Ignition - Moves - ... GATE CS 2019 - Switching...

https://www.sanfoundry.com/compiler-question-answer/ifa-epdm-moves-1/

Compiler - Top-Down Parsing
 Compiler - Predictive Parsing
 Compiler - Bottom-Up Parsing1
 Compiler - Bottom-Up Parsing2
 Compiler - Sentinel Grammar1
 Compiler - Sentinel Grammar2
 Compiler - LR Parser - 1
 Compiler - LR Parser - 2
 Compiler - Parsing Table - 1
 Compiler - Parsing Table - 2
 Compiler - Derivations - 1
 Compiler - Derivations - 2
 Compiler - Translation Syntax1
 Compiler - Translation Syntax2
 Compiler - L - Definition - 1
 Compiler - L - Definition - 2
 Compiler - Generation - 1
 Compiler - Generation - 2
 Compiler - Statements - 1
 Compiler - Statements - 2
 Compiler - Programming - 2
 Compiler - Programming - 1
 Compiler - Increment - 1
 Compiler - Increment - 2
 Compiler - Array Reference-1
 Compiler - Array Reference-2
 Compiler - Switch Case - 1
 Compiler - Switch Case - 2
 Compiler - Management - 1
 Compiler - Management - 2
 Compiler - Storage Location-1
 Compiler - Storage Location-2

6. Convert the NFA to DFA

a)

b)

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Compiler - Top-Down Parsing
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 Compiler - L - Definition - 2
 Compiler - Generation - 1
 Compiler - Generation - 2
 Compiler - Statements - 1
 Compiler - Statements - 2
 Compiler - Programming - 2
 Compiler - Programming - 1
 Compiler - Increment - 1
 Compiler - Increment - 2
 Compiler - Array Reference-1
 Compiler - Array Reference-2
 Compiler - Switch Case - 1
 Compiler - Switch Case - 2
 Compiler - Management - 1
 Compiler - Management - 2
 Compiler - Storage Location-1
 Compiler - Storage Location-2

6. Convert the NFA to DFA

a)

b)

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Compiler - Definitions - 2
 Compiler - Translation Syntax1
 Compiler - Translation Syntax2
 Compiler - C - Definition - 1
 Compiler - C - Definition - 2
 Compiler - Generation - 1
 Compiler - Generation - 2
 Compiler - Statements - 1
 Compiler - Statements - 2
 Compiler - Programming - 2
 Compiler - Programming - 1
 Compiler - Locals - 1
 Compiler - Locals - 2
 Compiler - Array Reference - 1
 Compiler - Array Reference - 2
 Compiler - Switch Case - 1
 Compiler - Switch Case - 2
 Compiler - Management - 1
 Compiler - Management - 2
 Compiler - Storage Location - 1
 Compiler - Storage Location - 2
 Compiler - Symbol Table - 1
 Compiler - Symbol Table - 2
 Compiler - Storage Allocation - 1
 Compiler - Storage Allocation - 2
 Compiler - Activation Record - 1
 Compiler - Activation Record - 2
 Compiler - Static Allocation - 1
 Compiler - Static Allocation - 2
 Compiler - Stack Allocation - 1
 Compiler - Stack Allocation - 2
 Compiler - Error Handling - 1
 Compiler - Error Handling - 2

2

a)

b)

c) All of the mentioned
 d) None of the mentioned

[View Answer](#)

Answer: a
 Explanation: Initially Q is empty. Then since the initial state of the DFA is {0} , {0} is added to Q.
 Since $2(0, a) = \{1, 2\}$, $\{1, 2\}$ is added to Q and $(\{0\}, a) = \{1, 2\}$.
 Since $2(0, b) = \Phi$, Φ is added to Q and $(\{0\}, b) = \Phi$.
 At this point $Q = \{\{0\}, \{1, 2\}, \Phi\}$. Similarly $(\{1, 2\}, b) = \{1, 3\}$. Hence $\{1, 3\}$ is added to Q . Similarly $(\{1, 3\}, a) = \{1, 2\}$ and $(\{1, 3\}, b) = \Phi$. Thus there are no new states to be added to Q . Since the transitions from all states of Q have been computed and no more states are added to Q, the conversion process stops here.

c) All of the mentioned
 d) None of the mentioned
 View Answer

Answer: a
 Explanation: Initially Q is empty. Then since the initial state of the DFA is {0} , {0} is added to Q.
 Since $2(0, a) = \{1, 2\}$, $\{1, 2\}$ is added to Q and $(\{0\}, a) = \{1, 2\}$.
 Since $2(0, b) = \Phi$, Φ is added to Q and $(\{0\}, b) = \Phi$.
 At this point $Q = \{\{0\}, \{1, 2\}, \Phi\}$. Similarly $(\{1, 2\}, b) = \{1, 3\}$. Hence $\{1, 3\}$ is added to Q . Similarly $(\{1, 3\}, a) = \{1, 2\}$ and $(\{1, 3\}, b) = \Phi$. Thus there are no new states to be added to Q . Since the transitions from all states of Q have been computed and no more states are added to Q, the conversion process stops here.

1. Let S and T be language over $\{a,b\}$ represented by the regular expressions $(a+b^*)^*$ and $(a+b)^*$, respectively. Which of the following is true? (GATE CS 2000)

- (a) $S \subset T$ (S is a subset of T)
- (b) $T \subset S$ (T is a subset of S)
- (c) $S = T$
- (d) $S \cap T = \emptyset$

Answer: (c).

2) Which one of the following languages over the alphabet $\{0,1\}$ is described by the regular expression: $(0+1)^*0(0+1)^*0(0+1)^*$?

- (A) The set of all strings containing the substring 00.
- (B) The set of all strings containing at most two 0's.
- (C) The set of all strings containing at least two 0's.
- (D) The set of all strings that begin and end with either 0 or 1.

Answer (C)

The regular expression has two 0's surrounded by $(0+1)^*$ which means accepted strings must have at least 2 0's.

3) Let $L = \{w \in (0+1)^* \mid w \text{ has even number of 1s}\}$, i.e. L is the set of all bit strings with even number of 1s. Which one of the regular expression below represents L?

- (A) $(0^*10^*)^*$
- (B) $0^*(10^*10^*)^*$
- (C) $0^*(10^*1^*)^*0^*$
- (D) $0^*1(10^*1^*)^*10^*$

Answer (B)

Option (A) is incorrect because it cannot accept "110"

Option (C) is incorrect because it accept a string with single 1.

Option (D) is incorrect because it cannot accept 11101

4) Which of the following statement(s) is/are correct?

- (A) only X
- (B) only Y
- (C) only Z
- (D) all of the above.

Explanation:

X: It is incorrect. since, a language L and its compliment can be infinite.

Y: It is correct. since, if the language contains epsilon then its initial state must be final also, otherwise the DFA wont be able to accept epsilon.

Z: It is incorrect. since, every language accepted by NFA is also accepted by some DFA. Hence, NFA and DFA both are equivalent in power.

Option (B) is true.

5) Which one of the following regular expression describes the language over $\{a, b\}$ which consists of no pair of consecutive b's?

- (A) $(a^*baa^*)(b + \text{epsilon})$
- (B) $(a + ba)^*(b + \text{epsilon})$

- (C) $(a^*baa^*)(b + \text{epsilon}) + a^*$
 (D) $(a^*ba^*)(b + \text{epsilon}) + a^*(b + \text{epsilon})$

Explanation:

- (A) It is incorrect. since, it does not contains (a or epsilon).
- (B) It is correct. since, it contains (epsilon, a, b, ba, ab,), i.e., no pair of consecutive b's.
- (C) It is incorrect. since, it does not contains 'ab' or 'aab'.
- (D) It is incorrect. since, it contains 'bb', which is not allowed.

Option (B) is true.

6 What is the length of the shortest string not in the language over alphabet $\{0, 1\}$ for regular expression given below:

$$1^*(0 + 1)^*1^*$$

- (A) seven
 (B) five
 (C) six
 (D) four

Explanation:

Check each string generated over the alphabet $\{0, 1\}$ until you reach the shortest string which is not generated by the given regular expression.

In this case, smallest string which is not generated by the given regular expression is 0110, whose length is four.

so, option (D) is true.

7 Consider X and Y are two languages over alphabet $\{0, 1\}$ represented by regular expression $0^*(10^*)^*$ and $(0^* + 1^*)^*$ respectively. which of the following is true?

- (A) X is proper subset of Y
 (B) Y is proper subset of X
 (C) $X = Y$
 (D) none

Explanation:

Here,

$$\begin{aligned} L(X) &= 0^*(10^*)^* \\ &= \{\text{epsilon}, 0, 1, 10, 01, 00, 11, \dots\} \end{aligned}$$

And.

$$\begin{aligned} L(Y) &= (0^* + 1^*)^* \\ &= \{\epsilon, 0, 1, 10, 01, 00, 11, \dots\} \end{aligned}$$

So, both languages are equivalent to each other.

Option (C) is true.

The Applications of these Automata are given as follows:

<https://www.sanfoundry.com/compiler-questions-answers-nfa-epsilon-moves-1/>

8. $S \rightarrow aSa \mid bSb \mid a \mid b$; the language generated by the above grammar is the set of

- a) All palindromes.
- b) All odd length palindromes
- c) Strings beginning and ending with the same symbol
- d) All even length palindromes

View Answer

Answer: b

Explanation: The strings accepted by language are $\{a, b, aaa, bbb, aba, bab, \dots\}$. All the strings are odd length palindromes.

9 Which one of the following languages over the alphabet $\{0, 1\}$ is described by the regular expression: $(0+1)^*0(0+1)^*0(0+1)^*$?

- a) strings with the substring 00
- b) strings with at most two 0's
- c) strings with at least two 0's
- d) strings beginning and ending with either 0 or 1

View Answer

Answer: c

Explanation: The RE having 2 0's padded by $(0+1)^*$ which means accepted strings must have at least 2 0's.

10. Does epsilon ring makes any change in the automata

- a) Yes
- b) No

View Answer

Answer: b

Explanation: It adds nothing new to the automata.

11. Which of the technique can be used to prove that a language is non regular?

- a) Ardens theorem
- b) Pumping Lemma
- c) Ogden's Lemma
- d) None of the mentioned

View Answer

Answer: b

Explanation: We use the powerful technique called Pumping Lemma, for showing certain languages not to be regular. We use Ardens theorem to find out a regular expression out of a finite automaton.

12 Which of the following language regular?

- a) $\{a^i b^j | i \geq 0\}$
- b) $\{a^i b^j | 0 < i < 5\}$
- c) $\{a^i b^j | i \geq 1\}$
- d) None of the mentioned

View Answer

Answer: b

Explanation: Here, i has limits i.e. the language is finite, contains few elements and can be graphed using a deterministic finite automata. Thus, it is regular. Others can be proved non regular using Pumping lemma.

13. Which of the following are non regular?

- a) The set of strings in $\{a, b\}^*$ with an even number of b's
- b) The set of strings in $\{a, b, c\}^*$ where there is no c anywhere to the left of a
- c) The set of strings in $\{0, 1\}^*$ that encode, in binary, an integer w that is a multiple of 3. Interpret the empty strings e as the number 0.
- d) None of the mentioned

View Answer

Answer: d

Explanation: All of the given languages are regular and finite and thus, can be represented using respective deterministic finite automata. We can also use mealy or moore machine to represent remainders for option c.

14. If L is DFA-regular, L' is

- a) Non regular
- b) DFA-regular
- c) Non-finite
- d) None of the mentioned

View Answer

Answer: b

Explanation: This is a simple example of a closure property: a property saying that the set of DFA-regular languages is closed under certain operations.

If L is DFA-regular, L' is

- a) Non regular
- b) DFA-regular
- c) Non-finite
- d) None of the mentioned

View Answer

Answer: b

Explanation: This is a simple example of a closure property: a property saying that the set of DFA-regular languages is closed under certain operations.

15. Which of the following options is incorrect?

- a) A language L is regular if and only if $\sim L$ has finite number of equivalent classes.
- b) Let L be a regular language. If $\sim L$ has k equivalent classes, then any DFA that recognizes L must have atmost k states.
- c) A language L is NFA-regular if and only if it is DFA-regular.
- d) None of the mentioned

View Answer

Answer: b

Explanation: Let L be a regular language. If $\sim L$ has k equivalent classes, then any DFA that recognizes L must have atleast k states.

16 Myhill Nerode does the following:

- a) Minimization of DFA
- b) Tells us exactly when a language is regular
- c) Both (a) and (b)
- d) None of the mentioned

View Answer

Answer: c

Explanation: In automata theory, the Myhill Nerode theorem provides a necessary and sufficient condition for a language to be regular. The Myhill Nerode theorem can be used to show a language L is regular by proving that the number of equivalence classes of R_L (relation) is finite.

17. NDFAs were introduced by _____

- a) Michael O Rabin & Dana Scott
- b) Dan Brown
- c) Sun micro system Labs

d) SAP Labs
View Answer

Answer: a

Explanation: NFAs were introduced Dana Scott and Michael O. Rabin who also showed their equivalence to DFAs.

18 The regular languages are not closed under

- a) Concatenation
 - b) Union
 - c) Kleene star
 - d) Complement
- View Answer

Answer: d

Explanation: RE are closed under

- Union (cf. picture)
- Intersection
- Concatenation
- Negation
- Keene closure.

19. An NFA's transition function returns

- a) A Boolean value
 - b) A state
 - c) A set of states
 - d) An edge
- View Answer

Answer: c

Explanation: A transition function $\Delta: Q \times \Sigma \rightarrow P(Q)$. Where $P(Q)$ denotes the power set of Q ...

20. How many strings of length less than 4 contains the language described by the regular expression $(x+y)^*y(a+ab)^*$?

- a) 7
- b) 10
- c) 12
- d) 11

View Answer

Answer : d

Explanation : string of length 0 = 1

string of length 1 = 4

string of length 2 = 3

string of length 3 = 3

21. Which of the following is true?

- a) $(01)^*0 = 0(10)^*$
- b) $(0+1)^*0(0+1)^*1(0+1) = (0+1)^*01(0+1)^*$
- c) $(0+1)^*01(0+1)^*+1*0^* = (0+1)^*$
- d) All of the mentioned

View Answer

Answer : d

Explanation : None.

22. A language is regular if and only if

- a) accepted by DFA
- b) accepted by PDA
- c) accepted by LBA
- d) accepted by Turing machine

View Answer

Answer : a

Explanation : All of above machine can accept regular language but all string accepted by machine is regular only for DFA.

23. Let the class of language accepted by finite state machine be L_1 and the class of languages represented by regular expressions be L_2 then

- a) $L_1 < L_2$
- b) $L_1 \geq L_2$
- c) $L_1 \cup L_2 = .^*$
- d) $L_1 = L_2$

View Answer

Answer : d

Explanation : Finite state machine and regular expression have same power to express a language.

24. Which of the following is not a regular expression?

- a) $[(a+b)^*(aa+bb)]^*$
- b) $[(0+1)-(0b+a1)^*(a+b)]^*$
- c) $(01+11+10)^*$
- d) $(1+2+0)^*(1+2)^*$

View Answer

Answer : b

Explanation : Except b all are regular expression*

25. Regular expression are

- a) Type 0 language

- b) Type 1 language
- c) Type 2 language
- d) Type 3 language

View Answer

Answer : a

Explanation : According to Chomsky hierarchy .

26. Which of the following is true?

- a) Every subset of a regular set is regular
- b) Every finite subset of non-regular set is regular
- c) The union of two non regular set is not regular
- d) Infinite union of finite set is regular

View Answer

Answer : b

Explanation : None.

27. Regular expressions are closed under

- a) Union
- b) Intersection
- c) Kleen star
- d) All of the mentioned

View Answer

Answer : d

Explanation : According to definition of regular expression.

7. Myhill Nerode does the following:

- a) Minimization of DFA
- b) Tells us exactly when a language is regular
- c) Both (a) and (b)
- d) None of the mentioned

View Answer

Answer: c

Explanation: In automata theory, the Myhill Nerode theorem provides a necessary and sufficient condition for a language to be regular. The Myhill Nerode theorem can be used to show a language L is regular by proving that the number of equivalence classes of R_L (relation) is finite.

28. Which of the following strings do not belong the given regular expression?

- (a)*(a+cba)
- a) aa
- b) aaa

- c) acba
 - d) acbacba
- View Answer

Answer: d

Explanation: The string acbacba is unacceptable by the regular expression $(a)^*(a+cba)$.

2. Let L denotes the language generated by the grammar $S \rightarrow OSO/00$. Which of the following is true? (GATE CS 2000)

- (a) $L = O$
- (b) L is regular but not O
- (c) L is context free but not regular
- (d) L is not context free

Answer: (b)

Explanation: Please note that grammar itself is not regular but language L is regular as L can be represented using a regular grammar, for example $S \rightarrow S00/00$.

References:

http://en.wikipedia.org/wiki/Regular_grammar

3. Consider the following two statements:

S1: $\{0^{2n} \mid n \geq 1\}$ is a regular language

S2: $\{0^m 0^n 0^{(m+n)} \mid m \geq 1 \text{ and } n \geq 2\}$ is a regular language

Which of the following statements is correct? (GATE CS 2001)

- a) Only S1 is correct
- b) Only S2 is correct
- c) Both S1 and S2 are correct
- d) None of S1 and S2 is correct

Answer: (c)

Explanation:

S1 can be written as $(00)^n$ where $n \geq 1$. And S2 can be written as $(00)^{(m+n)}$ where $m \geq 2$ and $n \geq 1$. S2 can be further reduced to $(00)^x$ where $x \geq 3$.

We can easily write regular grammars for both S1 and S2.

G1 $\rightarrow$ G100/00 (For S1)

G2 $\rightarrow$ G200/000000 (For S2)

Ques-1: Which one of the following language is Regular?

- (A) $\{wxw^R \mid w, x \in (a+b)^+\}$
 (B) $\{wxw^R \mid w \in (a+b)^*, x \in \{a,b\}\}$
 (C) $\{ww^Rx \mid w, x \in (a+b)^+\}$
 (D) $\{ww^R \mid w \in (a+b)^*\}$

Explanation:

- (A) It is correct, since this language can form regular expression which is $\{a(a+b)^+a + b(a+b)^+b\}$, i.e., start and end with same symbol.
- (B) It is deterministic context free language since, string before and after 'x' are same so, it is matched.
- (C) It cannot be regular since, ww^R is done at first which requires comparison which cannot be done via finite automata.
- (D) It is also not regular since, comparison is required.

Option (A) is true.

Ques-4: Consider the following statements:

$$S1 = \{(a^n)^m \mid n = 0\}$$

$$S2 = \{a^n b^n \mid n \geq 1\} \cup \{a^n b^m \mid n \geq 1, m \geq 1\}$$

Which one of the following is regular?

- (A) only S1
 (B) only S2
 (C) both S1 and S2
 (D) none

Explanation:

Both given languages are regular. Option (C) is correct.

1) $S \rightarrow aSa \mid bSb \mid a \mid b$; The language generated by the above grammar over the alphabet $\{a,b\}$ is the set of

- (A) All palindromes.
 (B) All odd length palindromes.
 (C) Strings that begin and end with the same symbol
 (D) All even length palindromes.

Answer (B)

The strings accepted by language are $\{a, b, aaa, bbb, aba, bab, \dots\}$. All of these strings are odd length palindromes.

1) Let P be a regular language and Q be context-free language such that $Q \subseteq P$. (For example, let P be the language represented by the regular expression p^*q^* and Q be $\{p^n q^n \mid n \in \mathbb{N}\}$). Then which of the following is ALWAYS regular?

- (A) $P \cap Q$
- (B) $P - Q$
- (C) $\Sigma^* - P$
- (D) $\Sigma^* - Q$

Answer (C)

The expression $\Sigma^* - P$ represents complement of P which is a regular language. Complement of Regular languages is also regular. Then a DFA that accepts the complement of L, i.e. $\Sigma^* - L$, can be obtained by swapping its accepting states with its non-accepting states.

2) Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^* ?

-1) abaabaaabaa
-2) aaaabaaaa
-3) baaaaabaaaab
-4) baaaaabaa

- (A) 1, 2 and 3
- (B) 2, 3 and 4
- (C) 1, 2 and 4
- (D) 1, 3 and 4

Answer (C)

Any combination of strings in set $\{ab, aa, baa\}$ will be in L^* .

-1) "abaabaaabaa" can be partitioned as a combination of strings in set $\{ab, aa, baa\}$. The partitions are "ab aa baa ab aa"
-2) "aaaabaaaa" can be partitioned as a combination of strings in set $\{ab, aa, baa\}$. The partitions are "aa ab aa aa"
-3) "baaaaabaaaab" cannot be partitioned as a combination of strings in set $\{ab, aa, baa\}$
-4) "baaaaabaa" can be partitioned as a combination of strings in set $\{ab, aa, baa\}$. The partitions are "baa aa ab aa"

3. Which of the following relates to Chomsky hierarchy?

- a) Regular < CFL < CSL < Unrestricted
- b) CFL < CSL < Unrestricted < Regular
- c) CSL < Unrestricted < CF < Regular
- d) None of the mentioned

View Answer

Answer: a

Explanation: The chomsky hierarchy lays down the following order:

Regular < CFL < CSL < Unrestricted

8. Which of the following strings is not generated by the given grammar:

$S \rightarrow SaSbS | e$

- a) aabb
- b) abab

- c) abaabb
 - d) None of the mentioned
- View Answer

Answer: d

Explanation: All the given options are generated by the given grammar. Using the methods of left and right derivations, it is simpler to look for string which a grammar can generate.

If L and L' are recursively enumerable, then L is

- A Regular
- B context-free
- context-sensitive

. The class of recursively enumerable language is known as:

- a) Turing Class
- b) Recursive Languages
- c) Universal Languages
- d) RE

View Answer

Answer: d

Explanation: RE or recursively enumerable is only called the class of recursively enumerable language.

. Which of the following statements are false?

- a) Every recursive language is recursively enumerable
- b) Recursively enumerable language may not be recursive
- c) Recursive languages may not be recursively enumerable
- d) None of the mentioned

View Answer

Answer: c

Explanation: Every recursive language is recursively enumerable but there exists recursively enumerable languages that are not recursive. If L is accepted by a Non deterministic TM T, and every possible sequence of moves of T causes it to halt, then L is recursive.

4. Choose the correct option:

Statement: If L1 and L2 are recursively enumerable languages over S, then the following is/are recursively enumerable.

- a) $L1 \cup L2$
- b) $L2 \cap L2$
- c) Both (a) and (b)
- d) None of the mentioned

View Answer

Answer: c

Explanation: Both the union and intersection operations preserve the property of recursive enumerability(Theorem).

5. If L is a recursive language, L' is:

- a) Recursive
- b) Recursively Enumerable
- c) Both (a) and (b)
- d) None of the mentioned

View Answer

Answer: c

Explanation: If T is a turing machine recognizing L, we can make it recognize L' by interchanging the two outputs. And every recursive language is recursively enumerable.

6. Choose the appropriate option:

Statement: If a language L is recursive, it is closed under the following operations:

- a) Union
- b) Intersection
- c) Complement
- d) All of the mentioned

View Answer

Answer: d

Explanation: The closure property of recursive languages include union, intersection and complement operations.

. A recursively enumerable language L can be recursive if:

- a) L' is recursively enumerable
- b) Every possible sequence of moves of T, the TM which accept L, causes it to halt
- c) Both (a) and (b)
- d) None of the mentioned

View Answer

Answer: c

Explanation: Theorem- If L is a recursively enumerable language whose complement is recursively enumerable, then L is recursive.

3. A language is regular if and only if

- a) accepted by DFA
- b) accepted by PDA
- c) accepted by LBA

d) accepted by Turing machine

[View Answer](#)

Answer : a

Explanation : All of above machine can accept regular language but all string accepted by machine is regular only for DFA.

4. Regular grammar is

- a) context free grammar
- b) non context free grammar
- c) english grammar
- d) none of the mentioned

[View Answer](#)

Answer : a

Explanation : Regular grammar is subset of context free grammar.

5. Let the class of language accepted by finite state machine be L_1 and the class of languages represented by regular expressions be L_2 then

- a) $L_1 < L_2$
- b) $L_1 \geq L_2$
- c) $L_1 \cup L_2 = .^*$
- d) $L_1 = L_2$

[View Answer](#)

Answer : d

Explanation : Finite state machine and regular expression have same power to express a language.

7. Regular expression are

- a) Type 0 language
- b) Type 1 language
- c) Type 2 language
- d) Type 3 language

[View Answer](#)

Answer : a

Explanation : According to Chomsky hierarchy

$S \rightarrow SS$ $S \rightarrow \lambda$ $S \rightarrow aSb$ $S \rightarrow bSa$ which type of grammar is it?

- a) Linear
- b) Nonlinear
- c) Both of the mentioned
- d) None of the mentioned

[View Answer](#)

Answer: a

Explanation: Grammar is non-linear because one of the rules (the first one) has two non-terminals on the right-hand side.

5. $S \rightarrow abS$ $S \rightarrow a$ is which grammar

- a) Right Linear Grammar
- b) Left Linear Grammar
- c) Right & Left Linear Grammar
- d) None of the mentioned

View Answer

Answer: a

Explanation: grammars in which all of the rules contain only one non-terminal on the left-hand side, and where in every case that non-terminal is the first symbol are called right Linear.

6. What are the two types of Linear Grammar?

- a) Right Linear
- b) Left Linear
- c) None of the mentioned
- d) Right & Left Linear

View Answer

Answer: d

Explanation: Linear grammar is of 2 types Left and Right Linear Grammar

8. Which Type of Grammar is it?

$S \rightarrow Aa$ $A \rightarrow Aab \mid \lambda$

- a) Right Linear
- b) Left Linear
- c) None of the mentioned
- d) Right & Left Linear

View Answer

Answer: b

Explanation: In this case they both correspond to the regular expression $(ab)^*a$

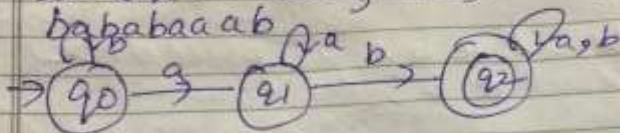
- Accepts a string a followed by $b\{a,b\}$

- Accepts a string containing even number of a's and any number of b's {a,b}
 - Accepts a string containing 2 consecutive a's followed by 2 consecutive b's {a,b}
 - Accepts a string that doesnot contain 3 consecutive a's {a,b}
-
- Accepts a string that contains even number of 0's and even number of 1's
 - Ends with one b and even number of a's
 - Accepts string aab
 - Accepts string that ends b not aa
 - Accepts a string that must not end withaa

Solutions

- ① Accepts a string a followed by b
 $\{a, b\}$ automata.

So we need to construct ~~any string~~
 with ab as substring
 it can be $ab, abb, bab, babab,$
 $bababab$



Hint: make sure your DFA must have
 at least 2 Transition for every
 state

like $\delta(q_0, b) \rightarrow q_0$
 $\delta(q_0, a) \rightarrow q_1$ for q_0

$\delta(q_1, a) \rightarrow q_1$
 $\delta(q_1, b) \rightarrow q_2$ for q_1

$\delta(q_2, a) \rightarrow q_2$
 $\delta(q_2, b) \rightarrow q_2$ for q_2

This is DFA, that doesn't
 contain any Null Transition
 But in NFA we can have
 Null Transition.